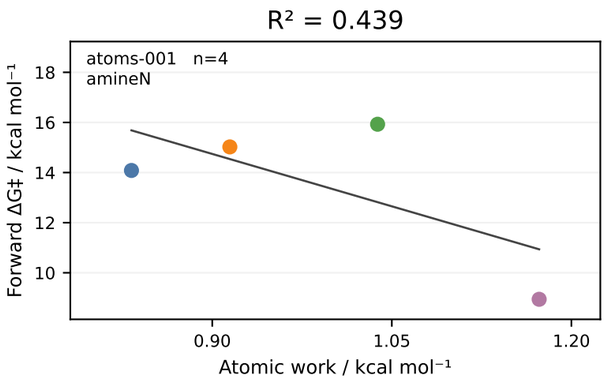
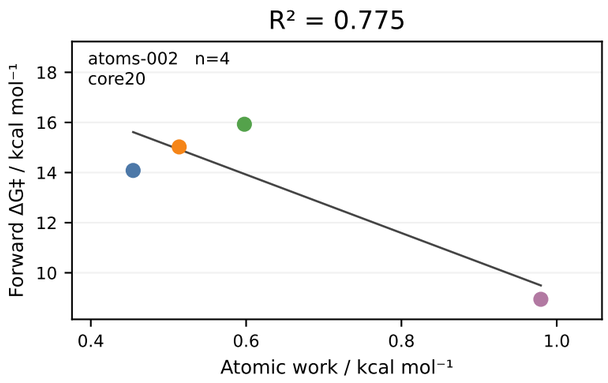
atoms | scatter | page 1 | C–N 2.50 → 2.10 Å

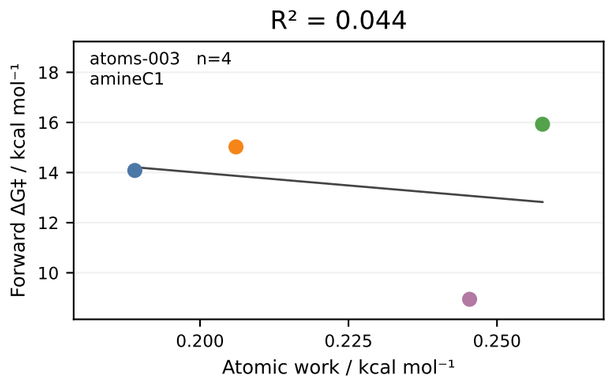
MoriPF RAFA14 RAFA15 RAFA16



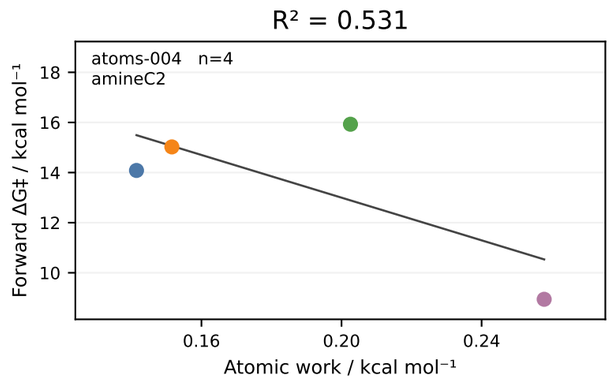
MoriPF:N45; R14:N43
R15:N43; R16:N43



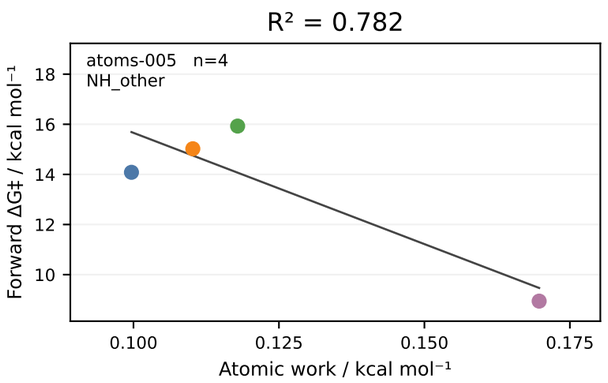
MoriPF:C20; R14:C20
R15:C20; R16:C20



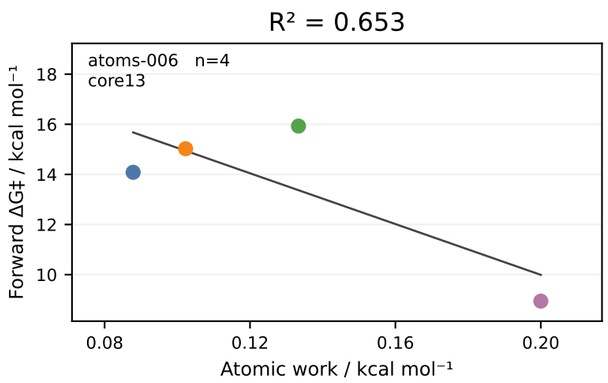
MoriPF:C48; R14:C46
R15:C46; R16:C46



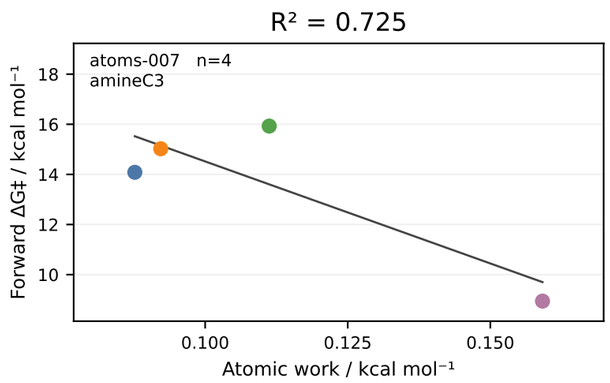
MoriPF:C51; R14:C49
R15:C49; R16:C49



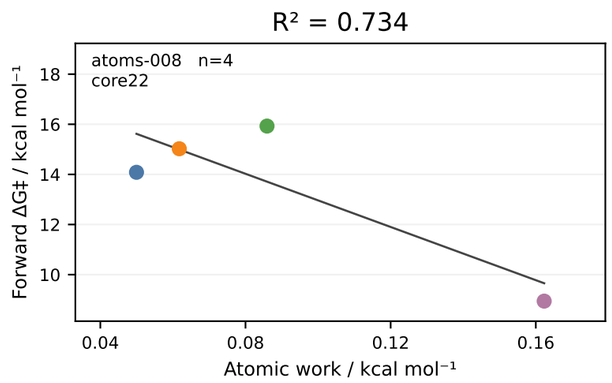
MoriPF:H47; R14:H45
R15:H45; R16:H45



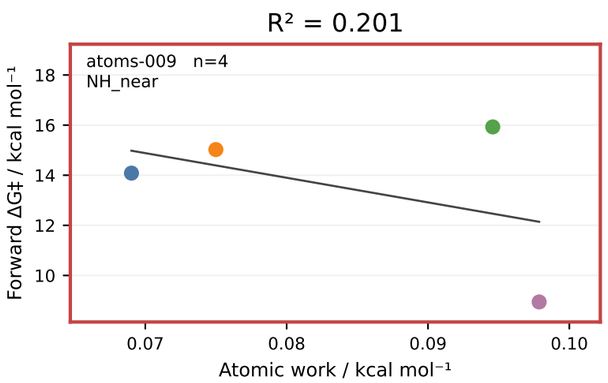
MoriPF:C13; R14:C13
R15:C13; R16:C13



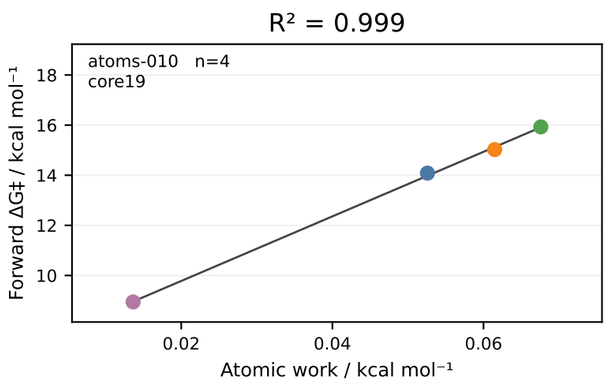
MoriPF:C54; R14:C52
R15:C52; R16:C52



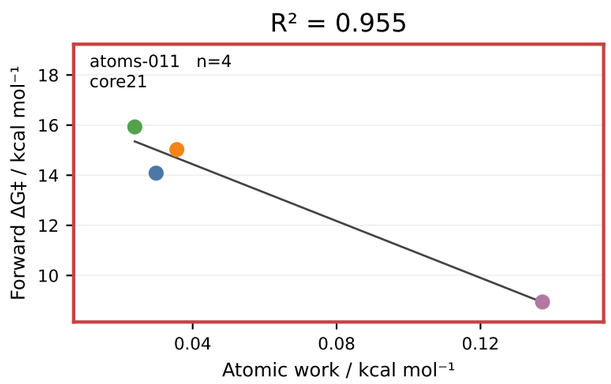
MoriPF:O22; R14:O22
R15:O22; R16:O22



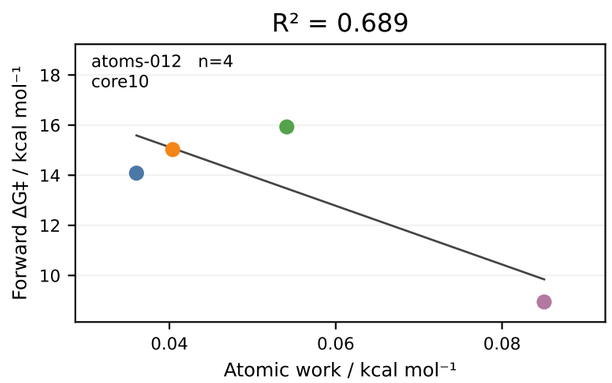
MoriPF:H46; R14:H44
R15:H44; R16:H44



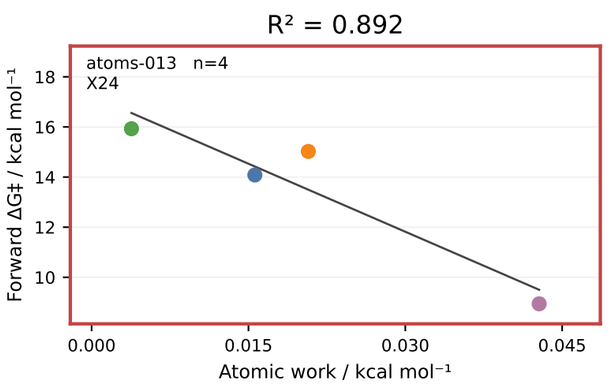
MoriPF:O19; R14:O19
R15:O19; R16:O19



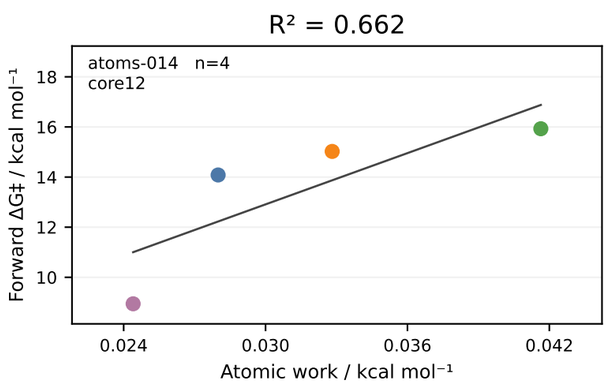
MoriPF:O21; R14:O21
R15:O21; R16:O21



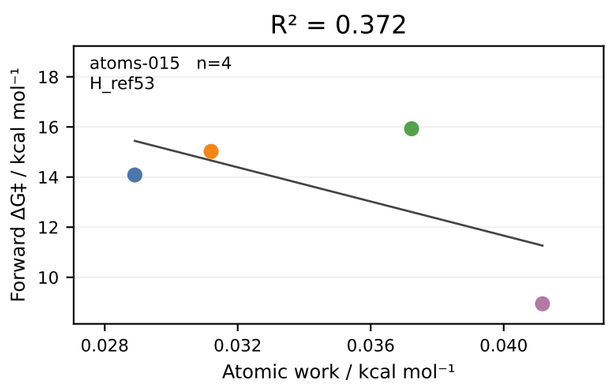
MoriPF:C10; R14:C10
R15:C10; R16:C10



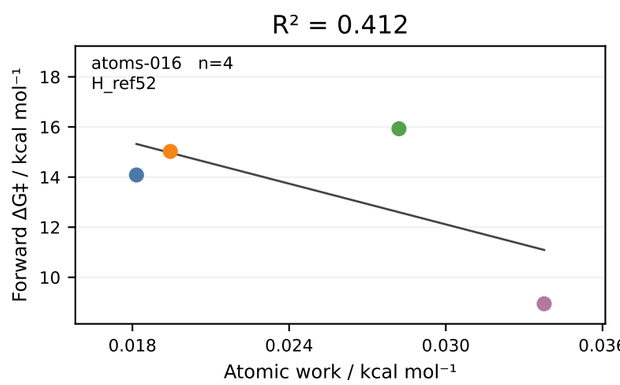
MoriPF:F33; R14:H57
R15:F31; R16:F31



MoriPF:C12; R14:C12
R15:C12; R16:C12



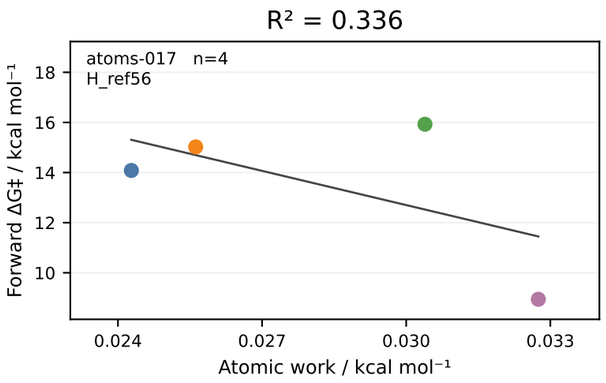
MoriPF:H53; R14:H51
R15:H51; R16:H51



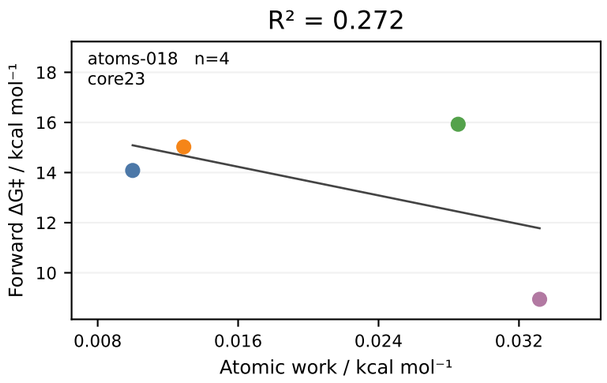
MoriPF:H52; R14:H50
R15:H50; R16:H50

atoms | scatter | page 2 | C–N 2.50 → 2.10 Å

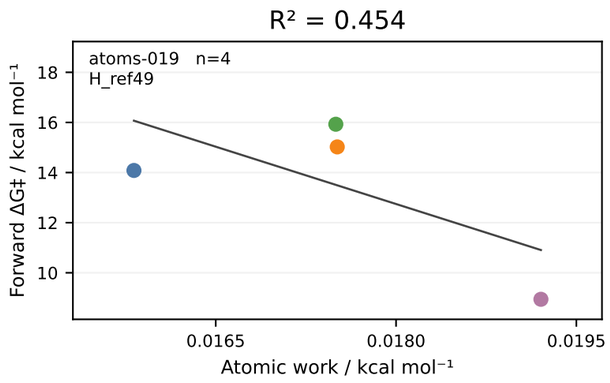
MoriPF RAFA14 RAFA15 RAFA16



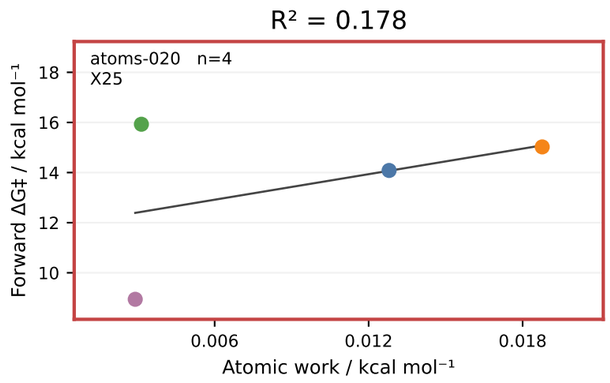
MoriPF:H56; R14:H54
R15:H54; R16:H54



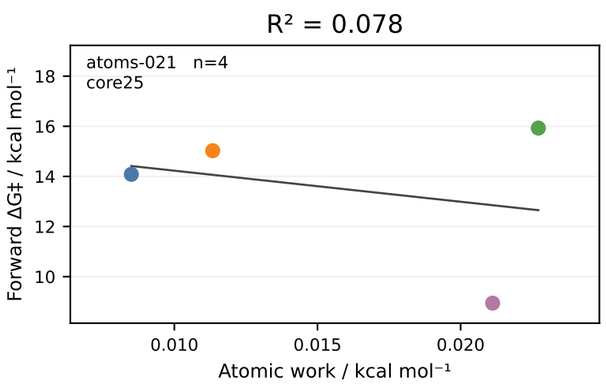
MoriPF:C23; R14:C23
R15:C23; R16:C23



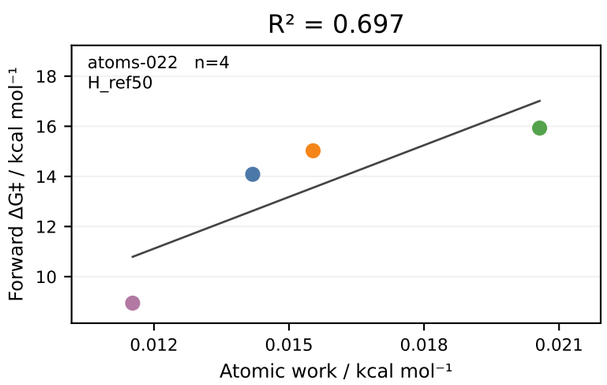
MoriPF:H49; R14:H47
R15:H47; R16:H47



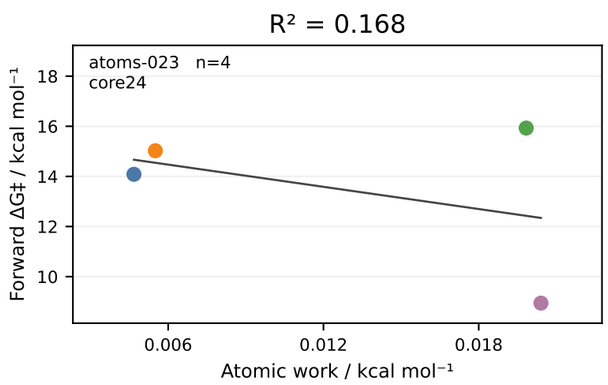
MoriPF:F29; R14:H56
R15:H56; R16:F29



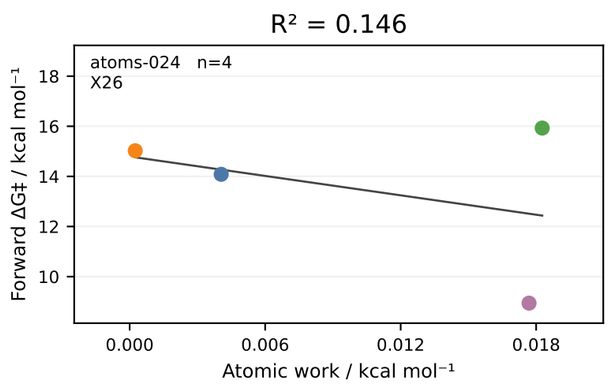
MoriPF:C25; R14:C25
R15:C25; R16:C25



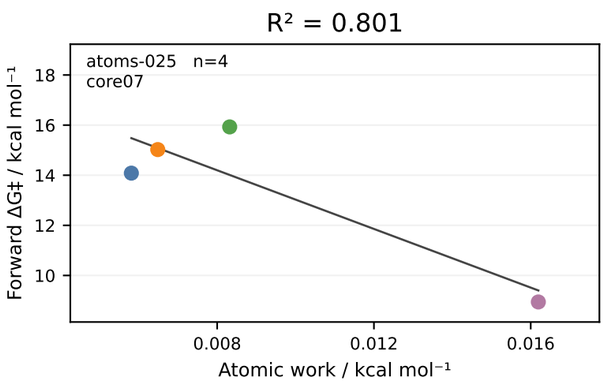
MoriPF:H50; R14:H48
R15:H48; R16:H48



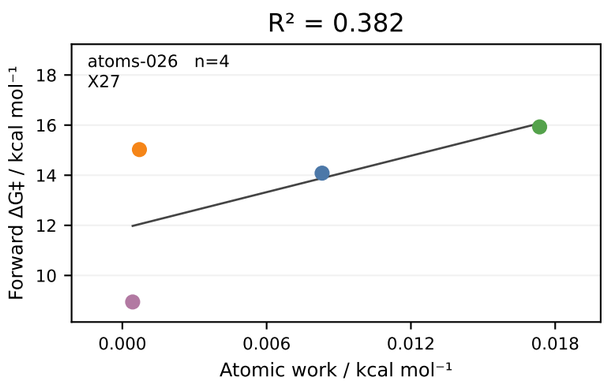
MoriPF:C24; R14:C24
R15:C24; R16:C24



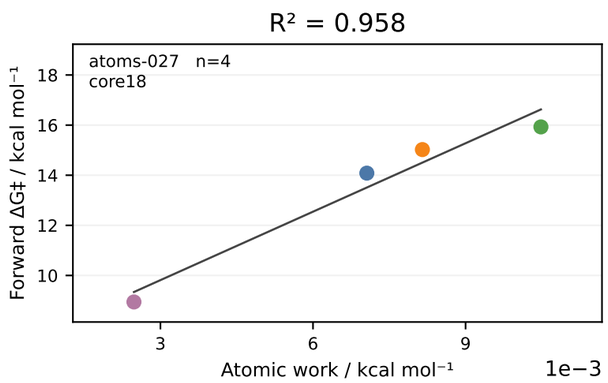
MoriPF:F32; R14:F31
R15:F30; R16:H56



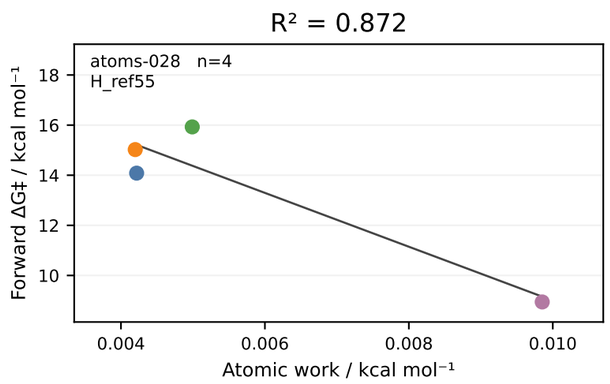
MoriPF:H7; R14:H7
R15:H7; R16:H7



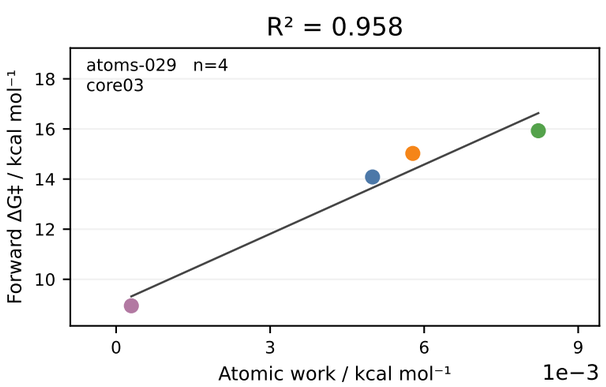
MoriPF:F30; R14:F29
R15:H57; R16:H57



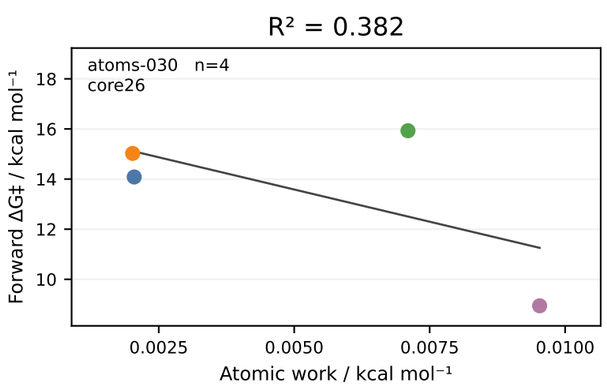
MoriPF:O18; R14:O18
R15:O18; R16:O18



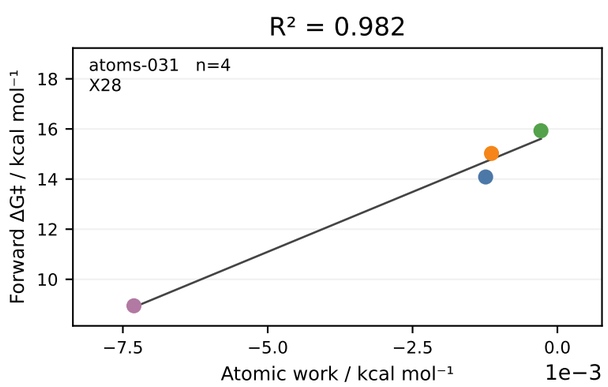
MoriPF:H55; R14:H53
R15:H53; R16:H53



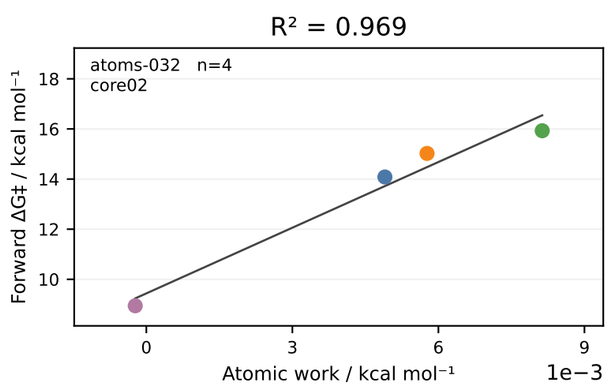
MoriPF:C3; R14:C3
R15:C3; R16:C3



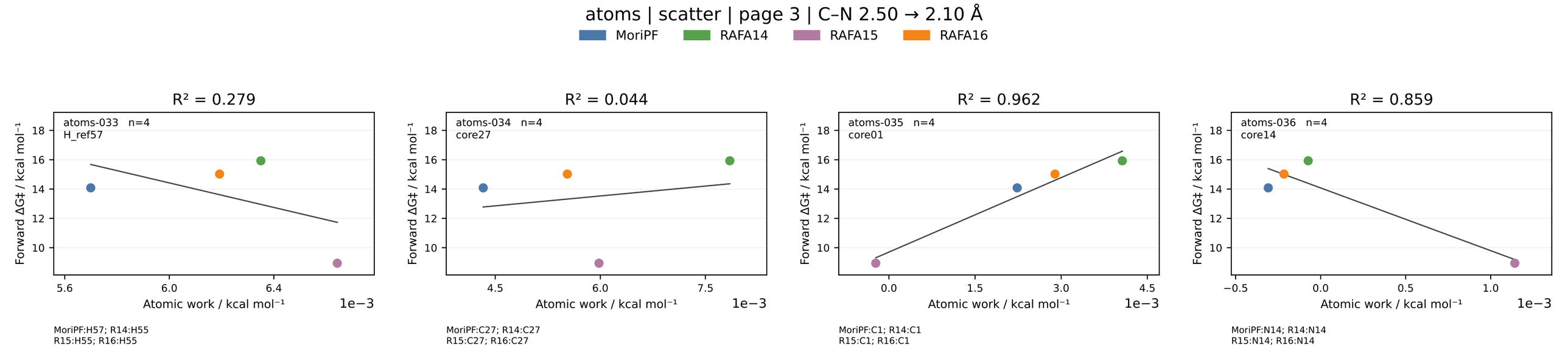
MoriPF:C26; R14:C26
R15:C26; R16:C26



MoriPF:F31; R14:F30
R15:F29; R16:F30

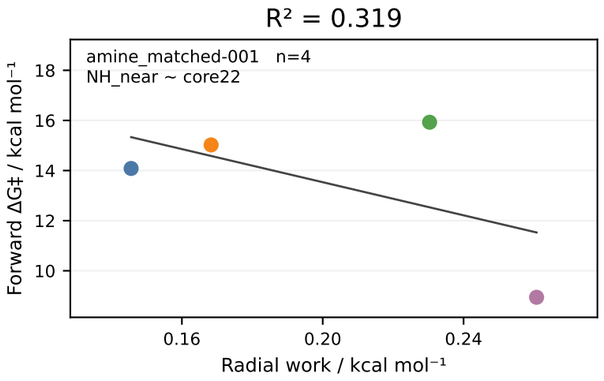


MoriPF:C2; R14:C2
R15:C2; R16:C2

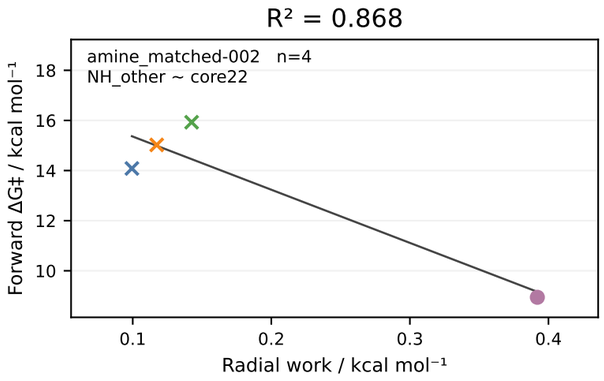

atoms | scatter | page 3 | C–N 2.50 → 2.10 Å
MoriPF
RAFA14
RAFA15
RAFA16
R² = 0.279
atoms-033 n=4
H_ref57
MoriPF:H57; R14:H55
R15:H55; R16:H55
R² = 0.044
atoms-034 n=4
core27
MoriPF:C27; R14:C27
R15:C27; R16:C27
R² = 0.962
atoms-035 n=4
core01
MoriPF:C1; R14:C1
R15:C1; R16:C1
R² = 0.859
atoms-036 n=4
core14
MoriPF:N14; R14:N14
R15:N14; R16:N14
R² = 0.866
atoms-037 n=4
core05
MoriPF:C5; R14:C5
R15:C5; R16:C5
R² = 0.959
atoms-038 n=4
core04
MoriPF:C4; R14:C4
R15:C4; R16:C4
R² = 0.946
atoms-039 n=4
core09
MoriPF:H9; R14:H9
R15:H9; R16:H9
R² = 0.562
atoms-040 n=4
sub_alkylC2
MoriPF:C38; R14:C36
R15:C36; R16:C36
Forward ΔG‡ / kcal mol⁻¹
Atomic work / kcal mol⁻¹
1e−3
1e−4

amine_matched | scatter | page 1 | C–N 2.50 → 2.10 Å

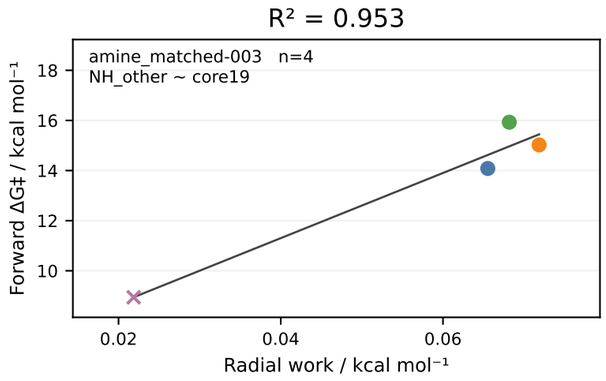
MoriPF RAFA14 RAFA15 RAFA16



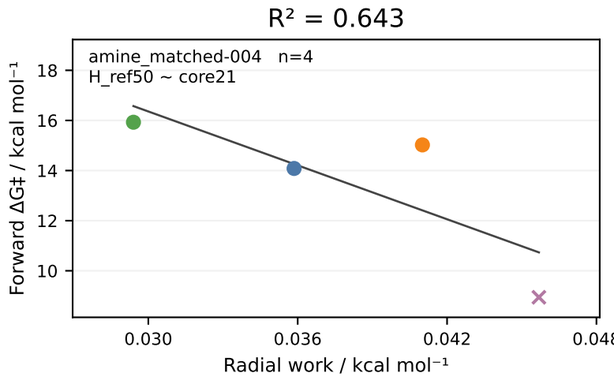
MoriPF:O22···H46; R14:O22···H44
R15:O22···H44; R16:O22···H44



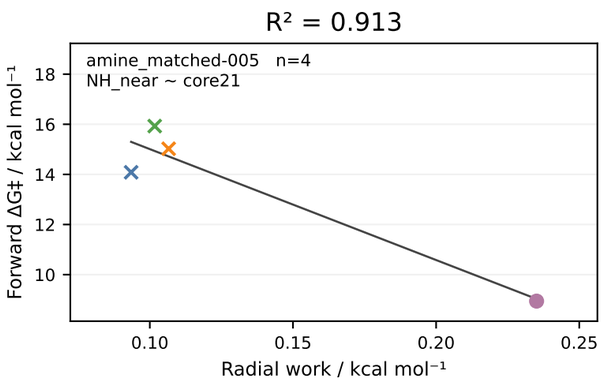
MoriPF:O22···H47; R14:O22···H45
R15:O22···H45; R16:O22···H45



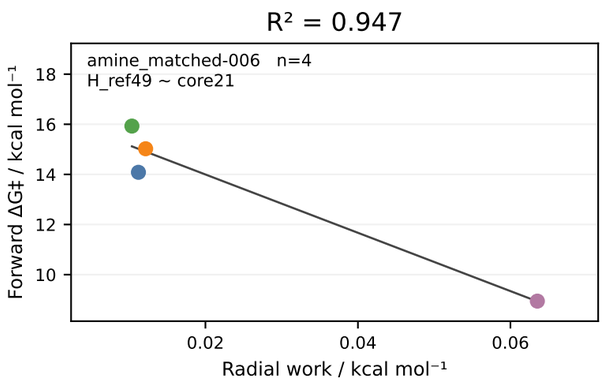
MoriPF:O19···H47; R14:O19···H45
R15:O19···H45; R16:O19···H45



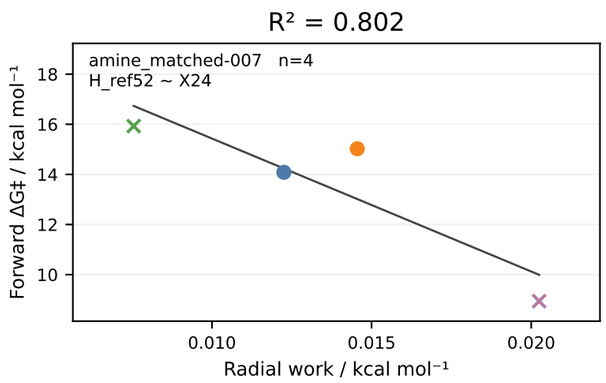
MoriPF:O21···H50; R14:O21···H48
R15:O21···H48; R16:O21···H48



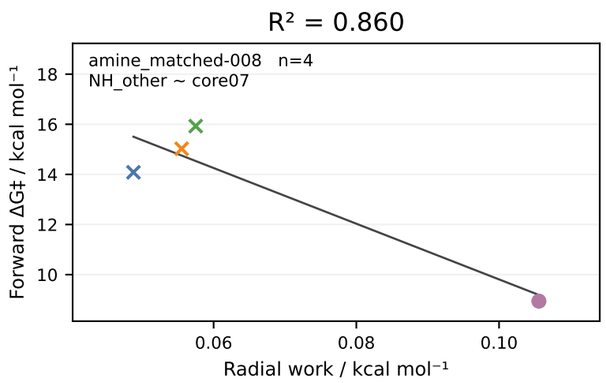
MoriPF:O21···H46; R14:O21···H44
R15:O21···H44; R16:O21···H44



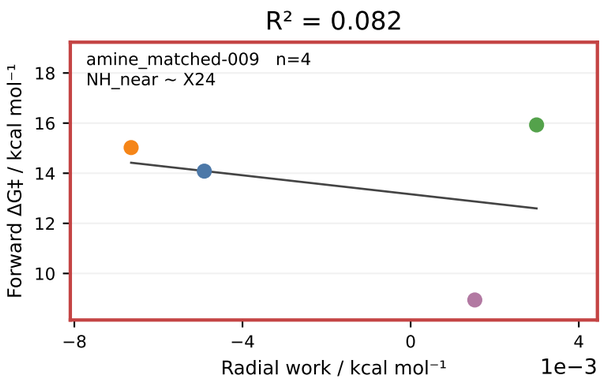
MoriPF:O21···H49; R14:O21···H47
R15:O21···H47; R16:O21···H47



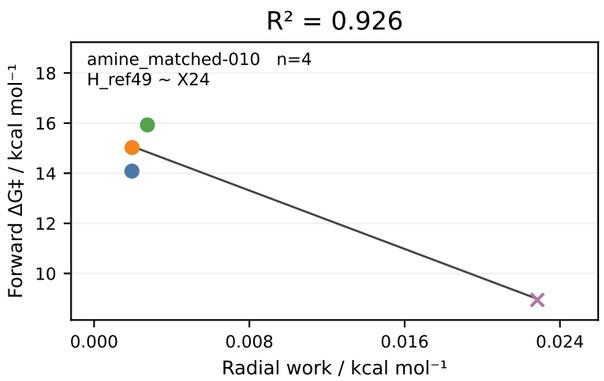
MoriPF:F33···H52; R14:H50···H57
R15:F31···H50; R16:F31···H50



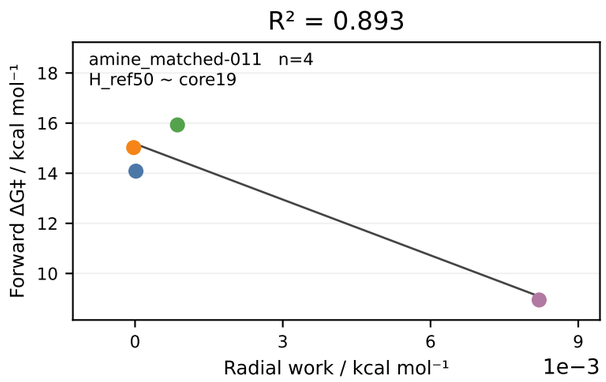
MoriPF:H7···H47; R14:H7···H45
R15:H7···H45; R16:H7···H45



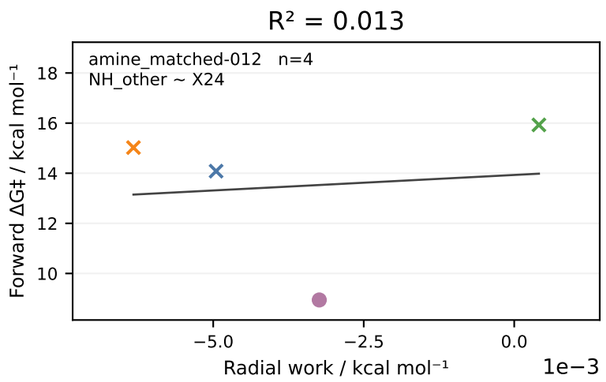
MoriPF:F33···H46; R14:H44···H57
R15:F31···H44; R16:F31···H44



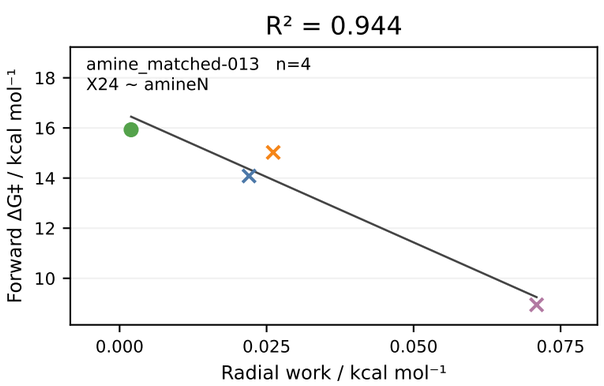
MoriPF:F33···H49; R14:H47···H57
R15:F31···H47; R16:F31···H47



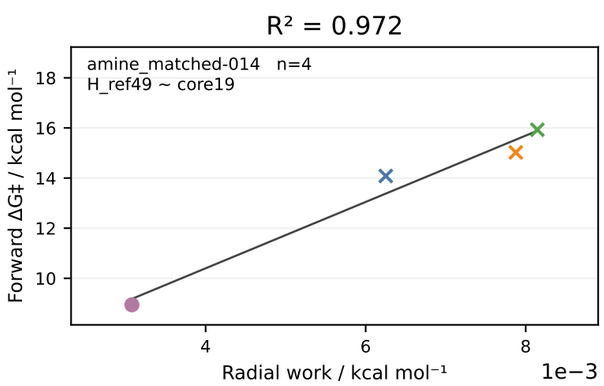
MoriPF:O19···H50; R14:O19···H48
R15:O19···H48; R16:O19···H48



MoriPF:F33···H47; R14:H45···H57
R15:F31···H45; R16:F31···H45



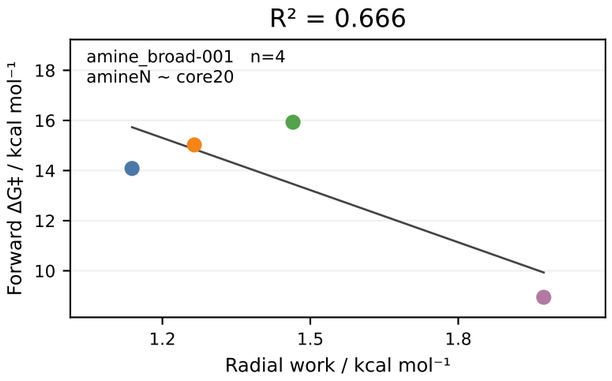
MoriPF:F33···N45; R14:N43···H57
R15:F31···N43; R16:F31···N43



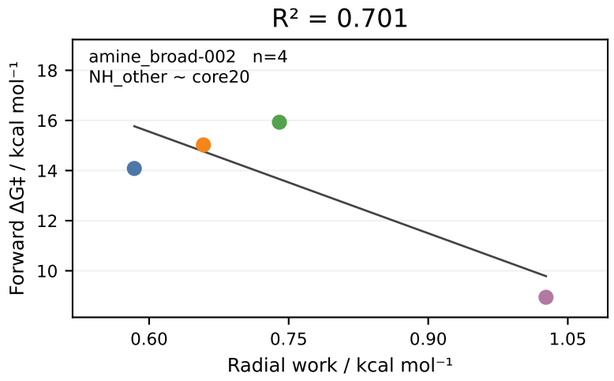
MoriPF:O19···H49; R14:O19···H47
R15:O19···H47; R16:O19···H47

amine_broad | scatter | page 1 | C–N 2.50 → 2.10 Å

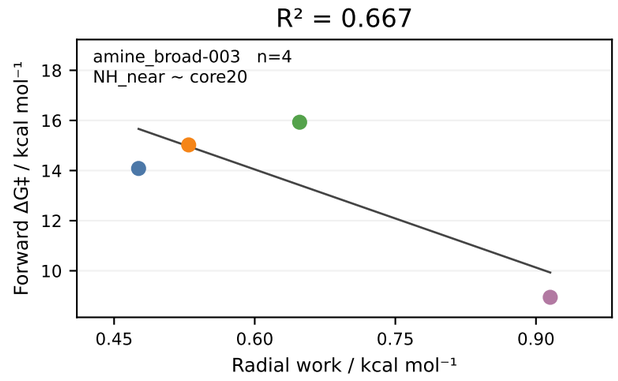
MoriPF RAFA14 RAFA15 RAFA16



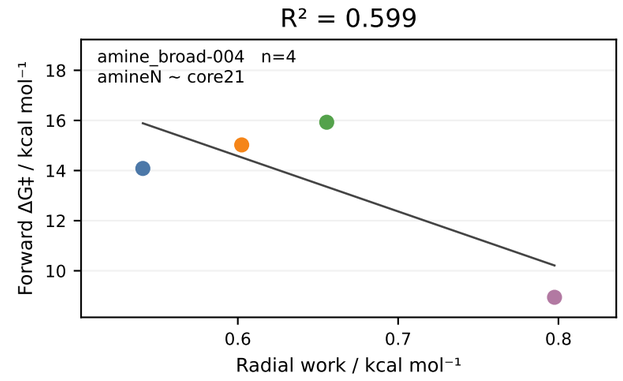
MoriPF:C20···N45; R14:C20···N43
R15:C20···N43; R16:C20···N43



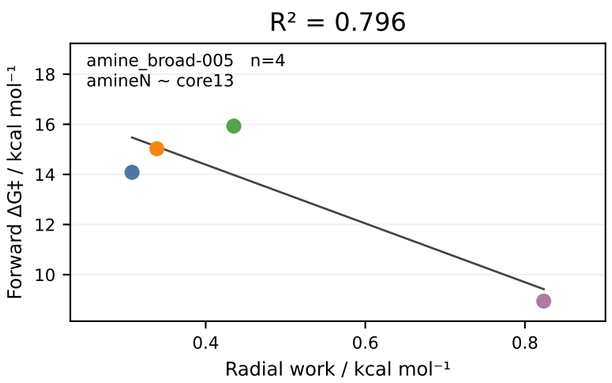
MoriPF:C20···H47; R14:C20···H45
R15:C20···H45; R16:C20···H45



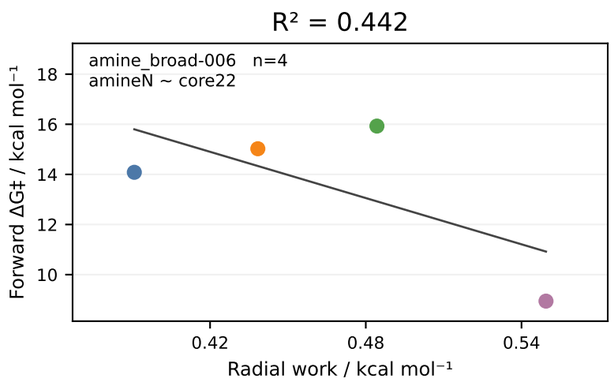
MoriPF:C20···H46; R14:C20···H44
R15:C20···H44; R16:C20···H44



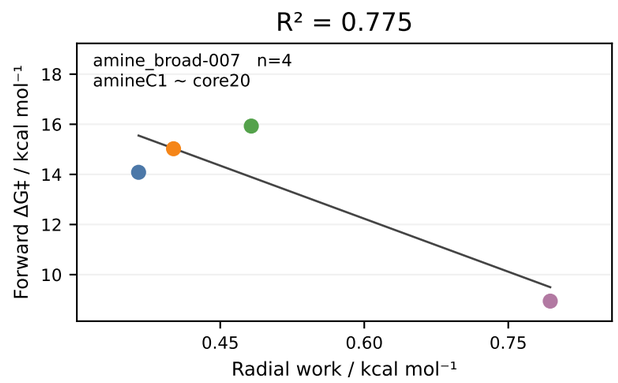
MoriPF:O21···N45; R14:O21···N43
R15:O21···N43; R16:O21···N43



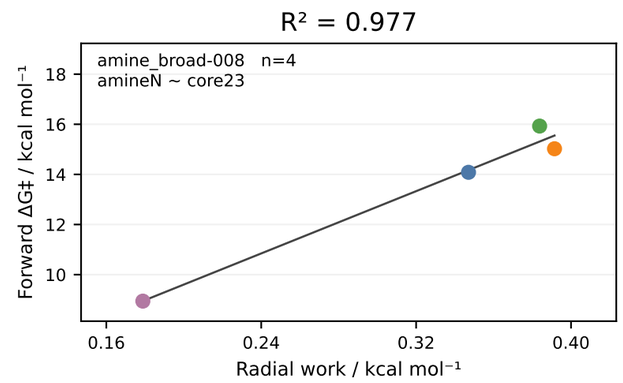
MoriPF:C13···N45; R14:C13···N43
R15:C13···N43; R16:C13···N43



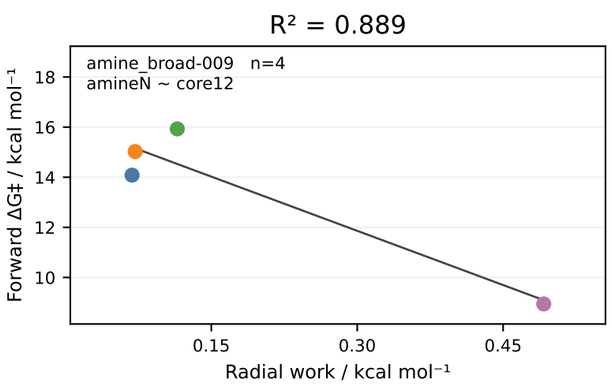
MoriPF:O22···N45; R14:O22···N43
R15:O22···N43; R16:O22···N43



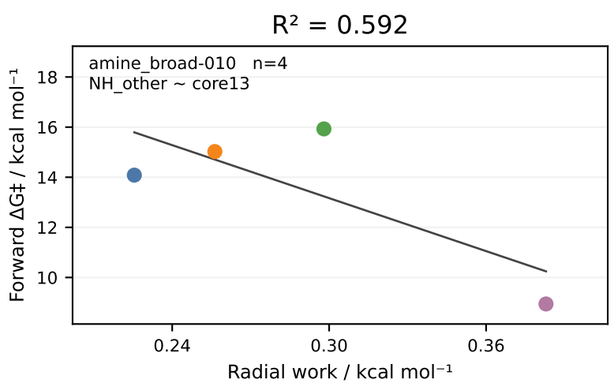
MoriPF:C20···C48; R14:C20···C46
R15:C20···C46; R16:C20···C46



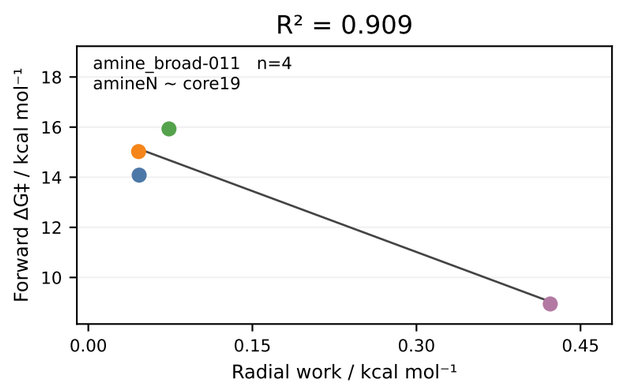
MoriPF:C23···N45; R14:C23···N43
R15:C23···N43; R16:C23···N43



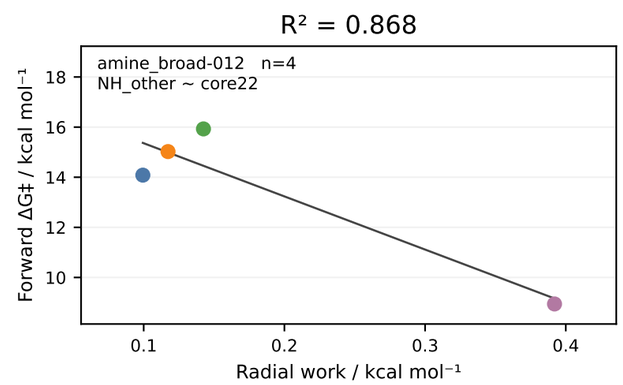
MoriPF:C12···N45; R14:C12···N43
R15:C12···N43; R16:C12···N43



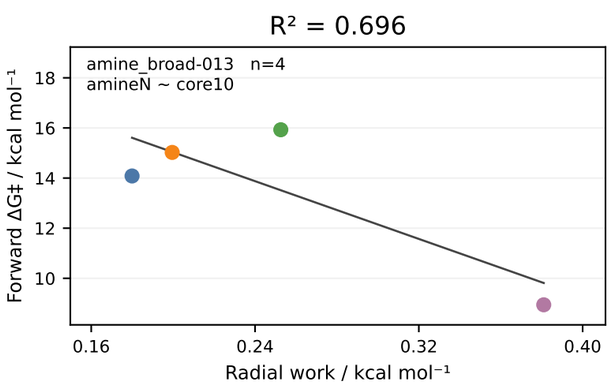
MoriPF:C13···H47; R14:C13···H45
R15:C13···H45; R16:C13···H45



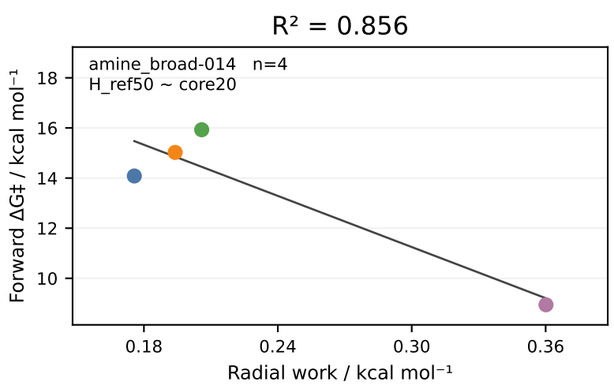
MoriPF:O19···N45; R14:O19···N43
R15:O19···N43; R16:O19···N43



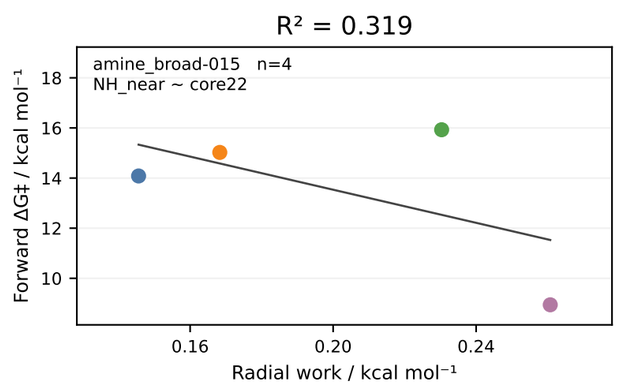
MoriPF:O22···H47; R14:O22···H45
R15:O22···H45; R16:O22···H45



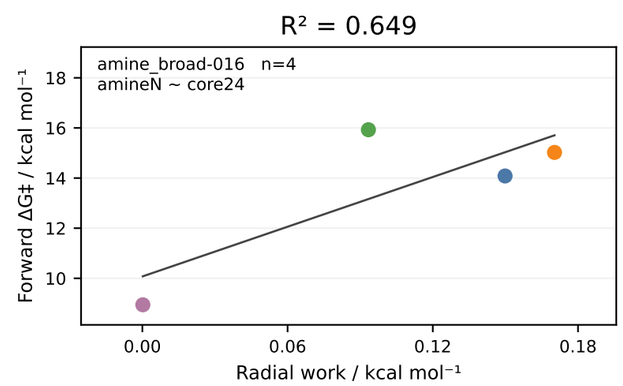
MoriPF:C10···N45; R14:C10···N43
R15:C10···N43; R16:C10···N43



MoriPF:C20···H50; R14:C20···H48
R15:C20···H48; R16:C20···H48



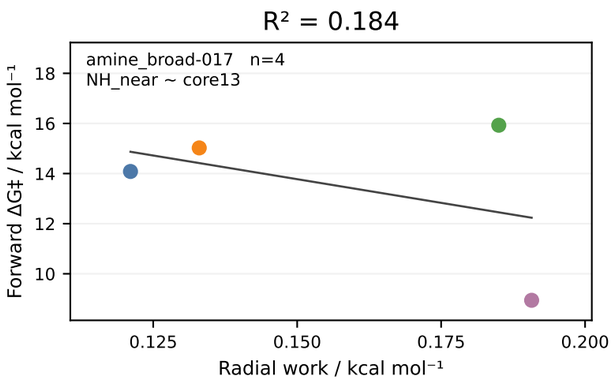
MoriPF:O22···H46; R14:O22···H44
R15:O22···H44; R16:O22···H44



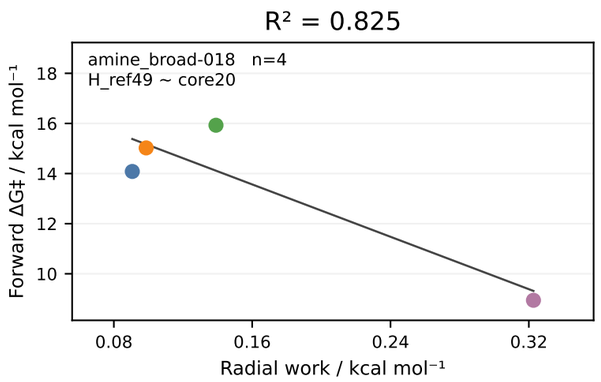
MoriPF:C24···N45; R14:C24···N43
R15:C24···N43; R16:C24···N43

amine_broad | scatter | page 2 | C–N 2.50 → 2.10 Å

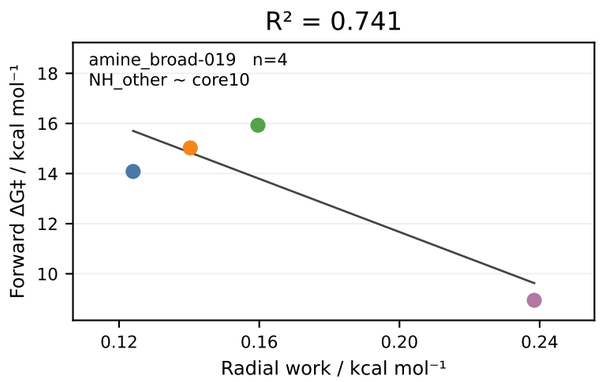
MoriPF RAFA14 RAFA15 RAFA16



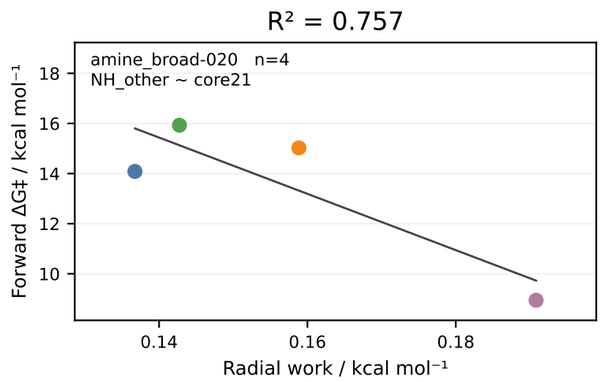
MoriPF:C13···H46; R14:C13···H44
R15:C13···H44; R16:C13···H44



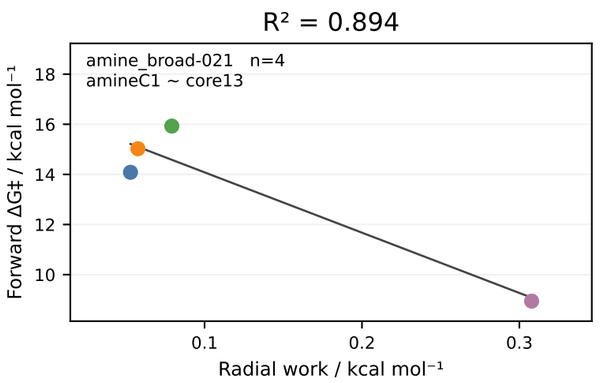
MoriPF:C20···H49; R14:C20···H47
R15:C20···H47; R16:C20···H47



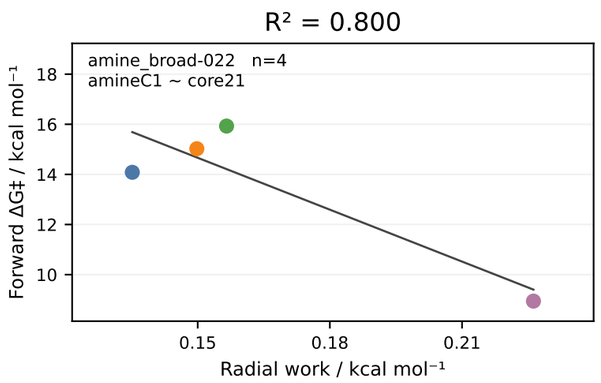
MoriPF:C10···H47; R14:C10···H45
R15:C10···H45; R16:C10···H45



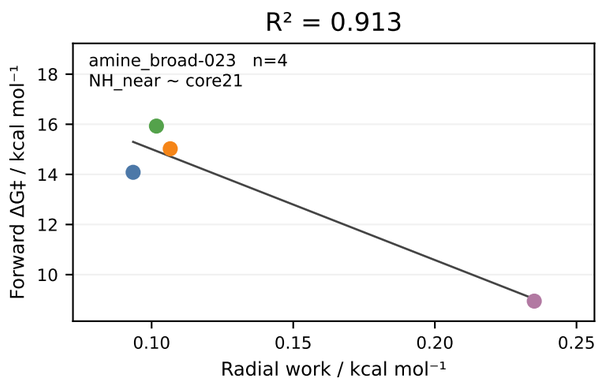
MoriPF:O21···H47; R14:O21···H45
R15:O21···H45; R16:O21···H45



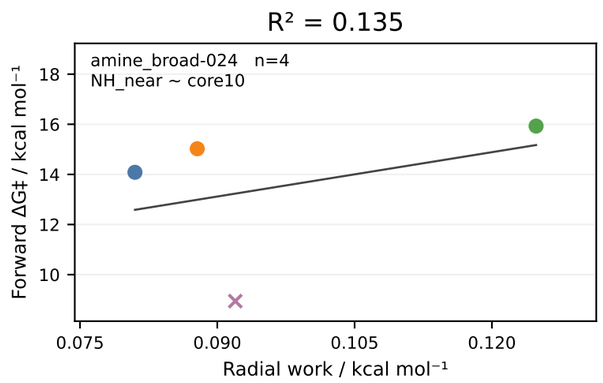
MoriPF:C13···C48; R14:C13···C46
R15:C13···C46; R16:C13···C46



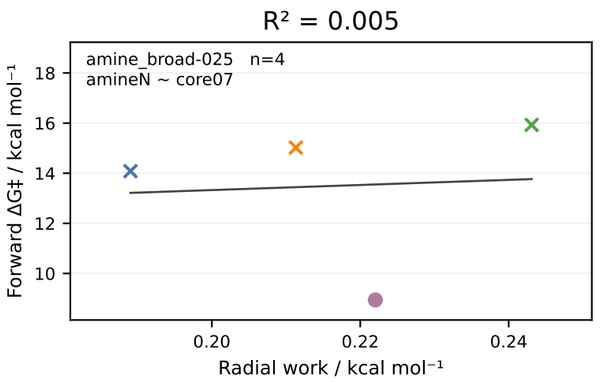
MoriPF:O21···C48; R14:O21···C46
R15:O21···C46; R16:O21···C46



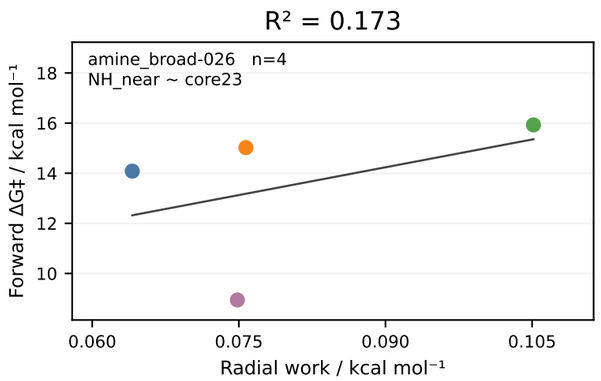
MoriPF:O21···H46; R14:O21···H44
R15:O21···H44; R16:O21···H44



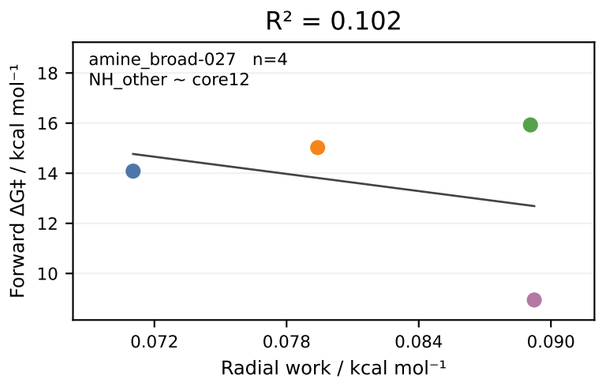
MoriPF:C10···H46; R14:C10···H44
R15:C10···H44; R16:C10···H44



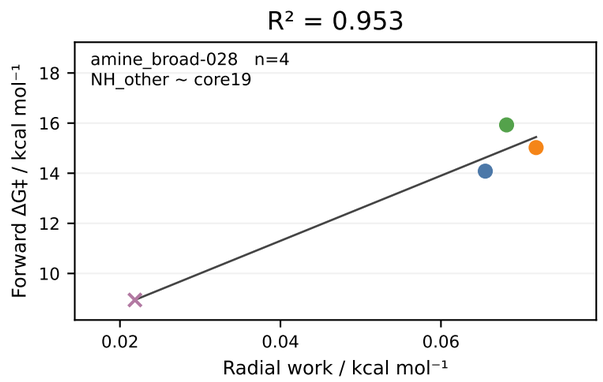
MoriPF:H7···N45; R14:H7···N43
R15:H7···N43; R16:H7···N43



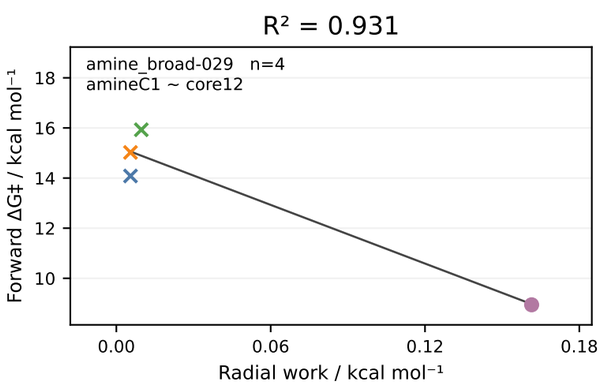
MoriPF:C23···H46; R14:C23···H44
R15:C23···H44; R16:C23···H44



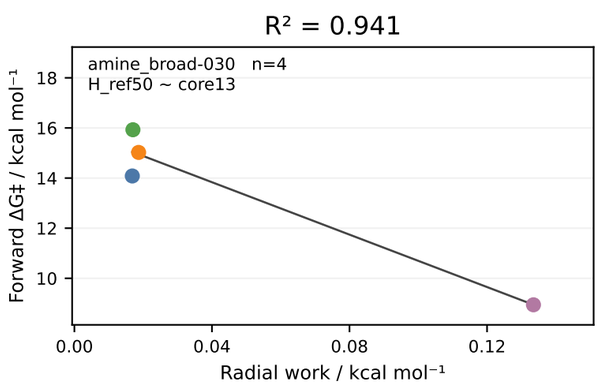
MoriPF:C12···H47; R14:C12···H45
R15:C12···H45; R16:C12···H45



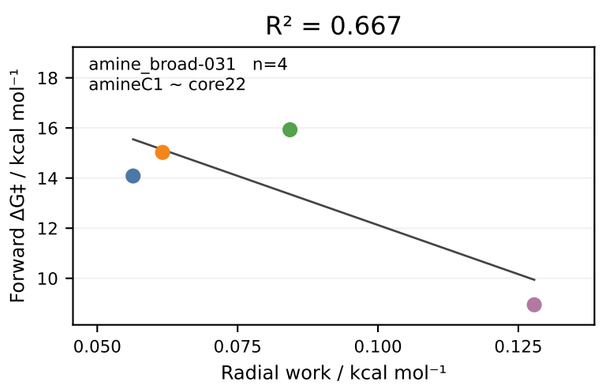
MoriPF:O19···H47; R14:O19···H45
R15:O19···H45; R16:O19···H45



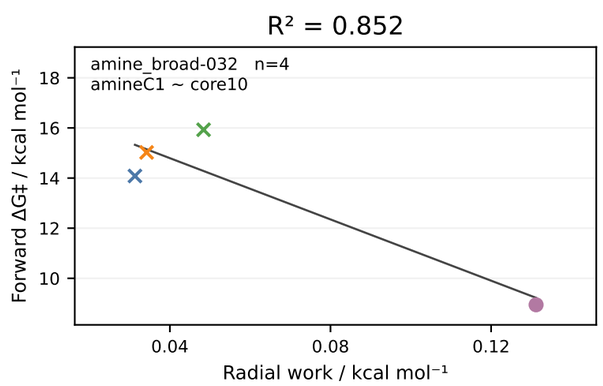
MoriPF:C12···C48; R14:C12···C46
R15:C12···C46; R16:C12···C46



MoriPF:C13···H50; R14:C13···H48
R15:C13···H48; R16:C13···H48



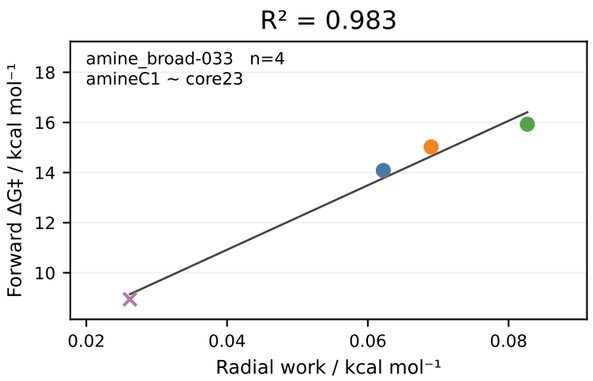
MoriPF:O22···C48; R14:O22···C46
R15:O22···C46; R16:O22···C46



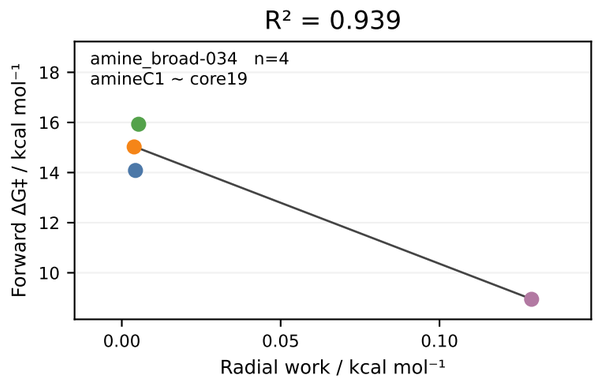
MoriPF:C10···C48; R14:C10···C46
R15:C10···C46; R16:C10···C46

amine_broad | scatter | page 3 | C–N 2.50 → 2.10 Å

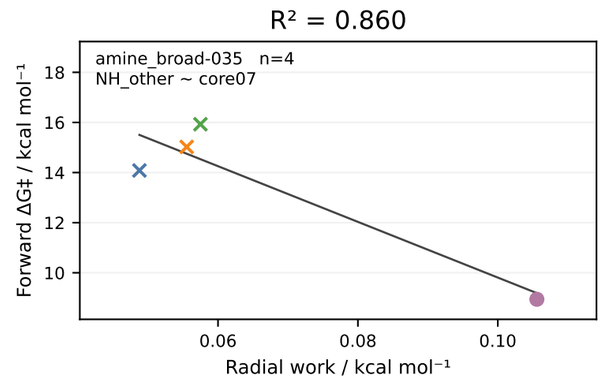
MoriPF RAFA14 RAFA15 RAFA16



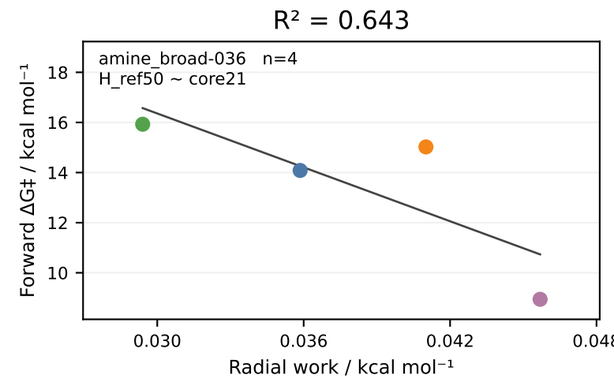
MoriPF:C23···C48; R14:C23···C46
R15:C23···C46; R16:C23···C46



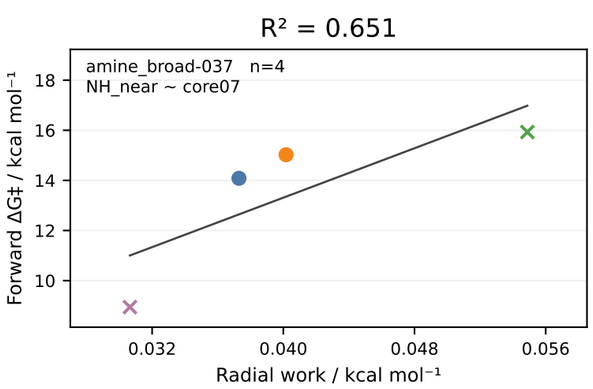
MoriPF:O19···C48; R14:O19···C46
R15:O19···C46; R16:O19···C46



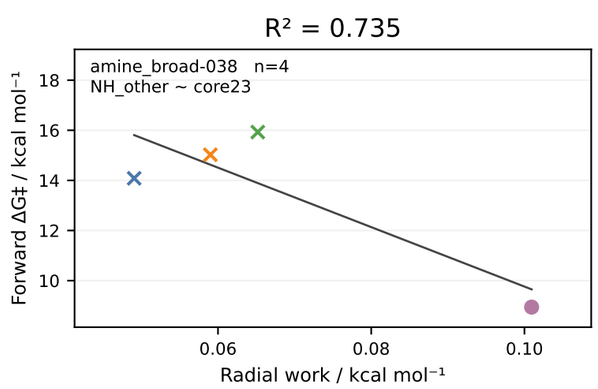
MoriPF:H7···H47; R14:H7···H45
R15:H7···H45; R16:H7···H45



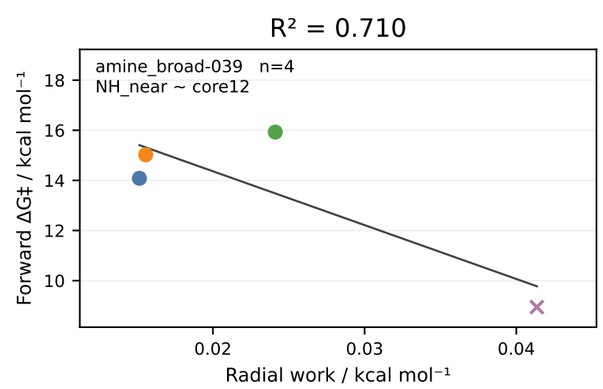
MoriPF:O21···H50; R14:O21···H48
R15:O21···H48; R16:O21···H48



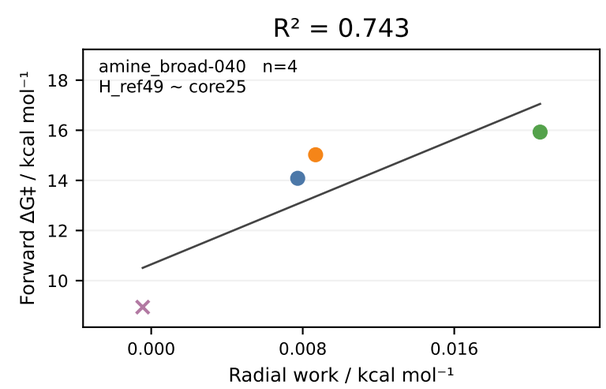
MoriPF:H7···H46; R14:H7···H44
R15:H7···H44; R16:H7···H44



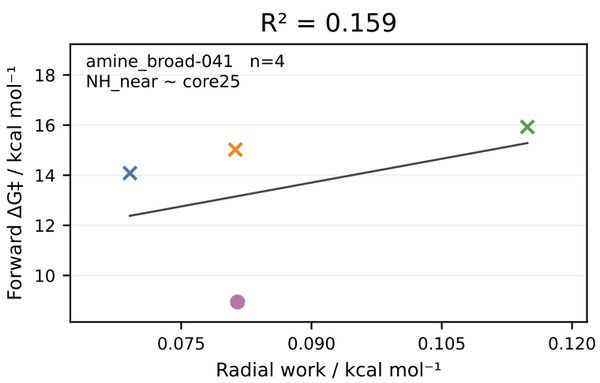
MoriPF:C23···H47; R14:C23···H45
R15:C23···H45; R16:C23···H45



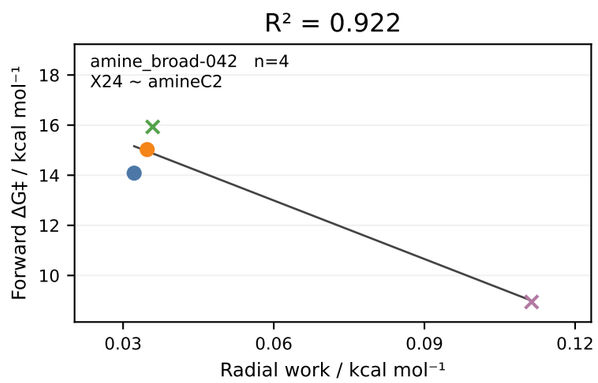
MoriPF:C12···H46; R14:C12···H44
R15:C12···H44; R16:C12···H44



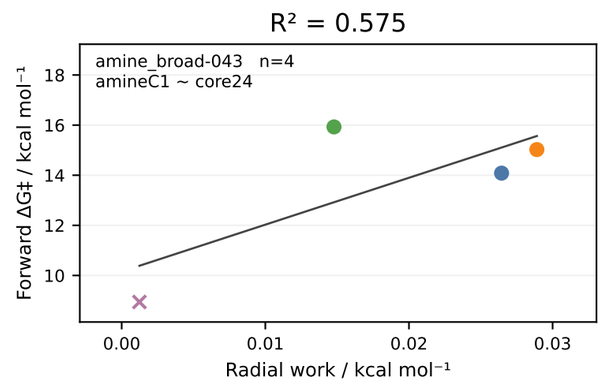
MoriPF:C25···H49; R14:C25···H47
R15:C25···H47; R16:C25···H47



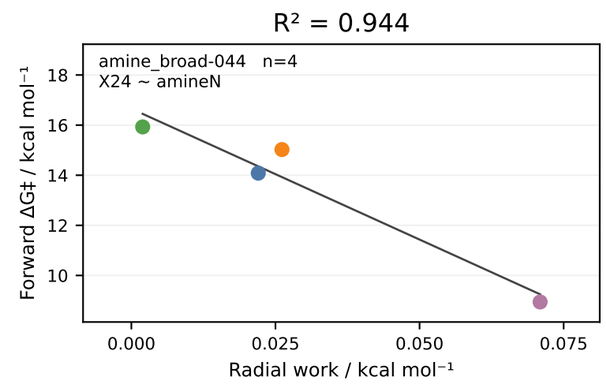
MoriPF:C25···H46; R14:C25···H44
R15:C25···H44; R16:C25···H44



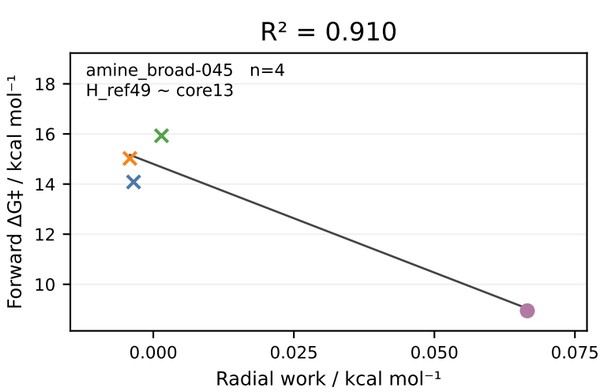
MoriPF:F33···C51; R14:C49···H57
R15:F31···C49; R16:F31···C49



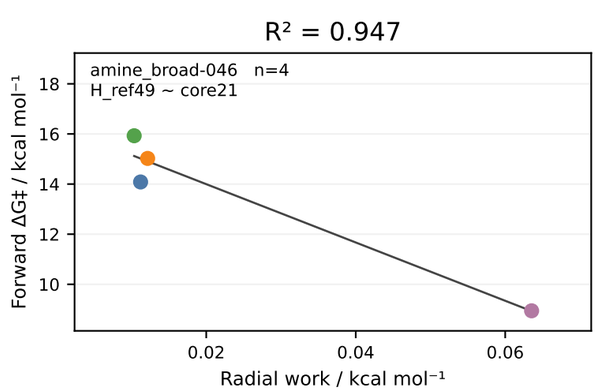
MoriPF:C24···C48; R14:C24···C46
R15:C24···C46; R16:C24···C46



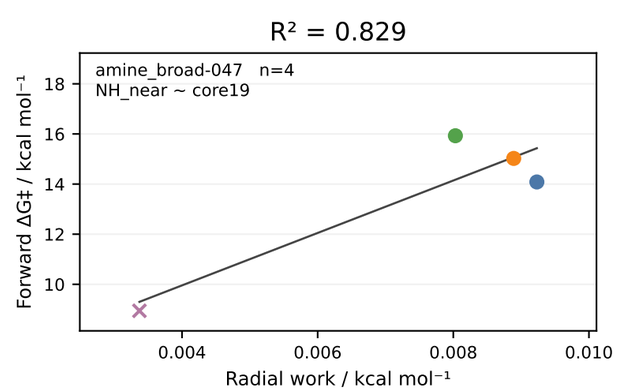
MoriPF:F33···N45; R14:N43···H57
R15:F31···N43; R16:F31···N43



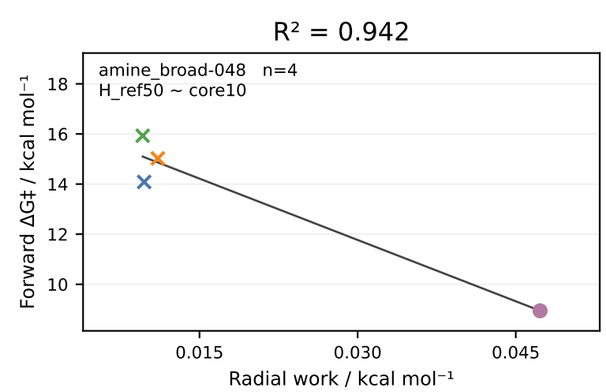
MoriPF:C13···H49; R14:C13···H47
R15:C13···H47; R16:C13···H47



MoriPF:O21···H49; R14:O21···H47
R15:O21···H47; R16:O21···H47



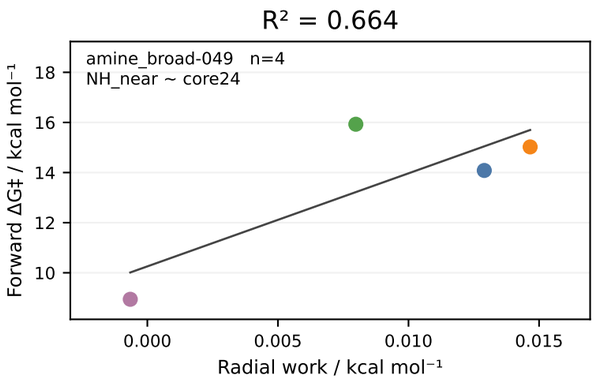
MoriPF:O19···H46; R14:O19···H44
R15:O19···H44; R16:O19···H44



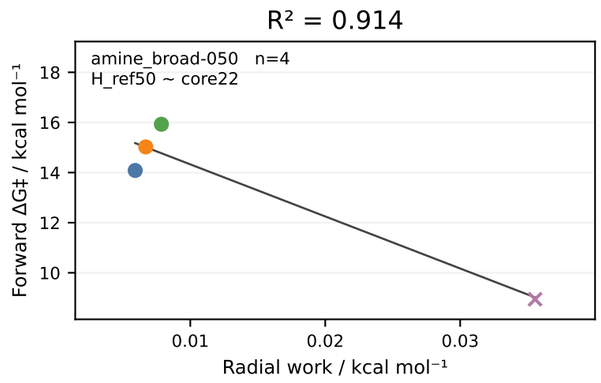
MoriPF:C10···H50; R14:C10···H48
R15:C10···H48; R16:C10···H48

amine_broad | scatter | page 4 | C–N 2.50 → 2.10 Å

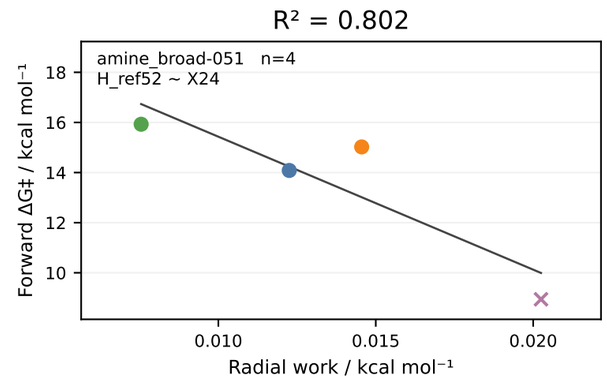
MoriPF RAFA14 RAFA15 RAFA16



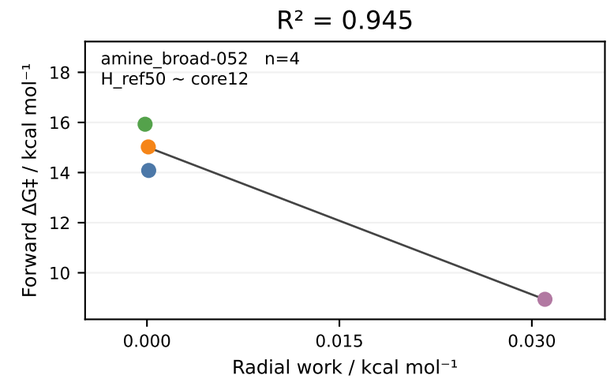
MoriPF:C24···H46; R14:C24···H44
R15:C24···H44; R16:C24···H44



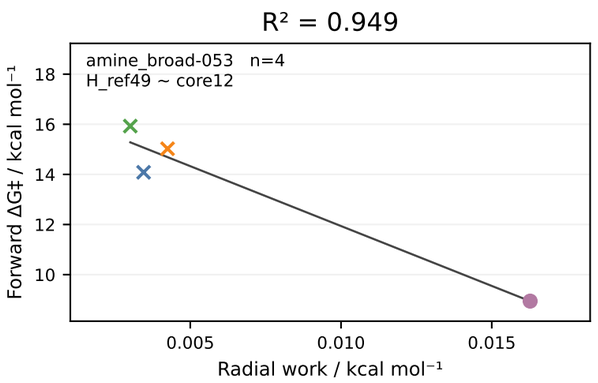
MoriPF:O22···H50; R14:O22···H48
R15:O22···H48; R16:O22···H48



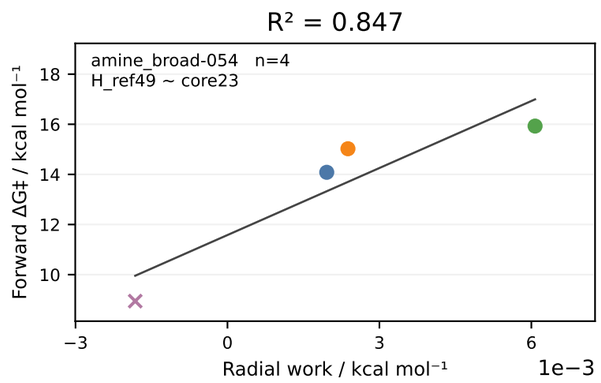
MoriPF:F33···H52; R14:H50···H57
R15:F31···H50; R16:F31···H50



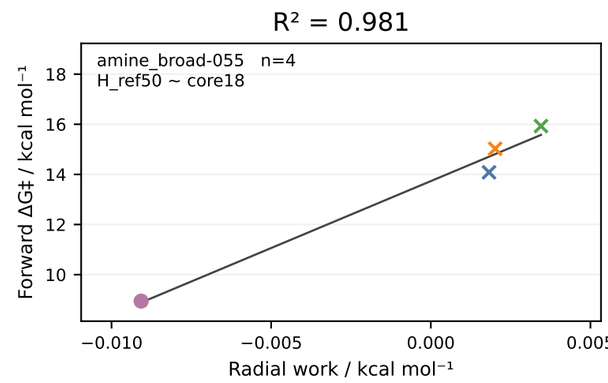
MoriPF:C12···H50; R14:C12···H48
R15:C12···H48; R16:C12···H48



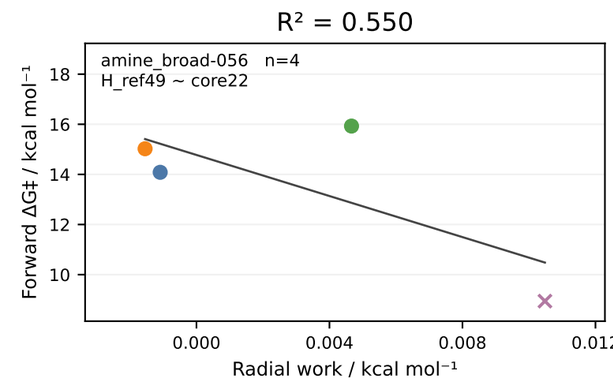
MoriPF:C12···H49; R14:C12···H47
R15:C12···H47; R16:C12···H47



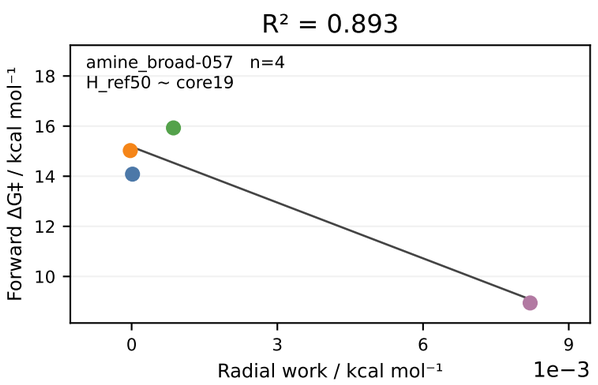
MoriPF:C23···H49; R14:C23···H47
R15:C23···H47; R16:C23···H47



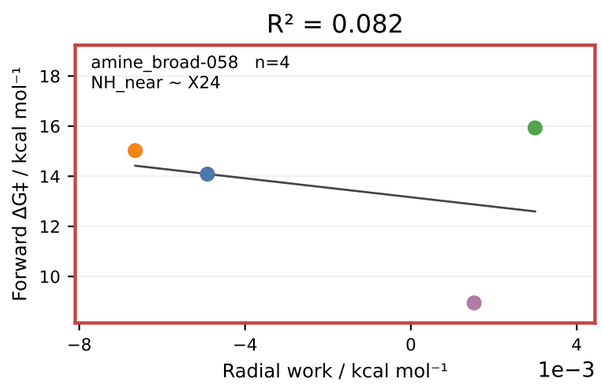
MoriPF:O18···H50; R14:O18···H48
R15:O18···H48; R16:O18···H48



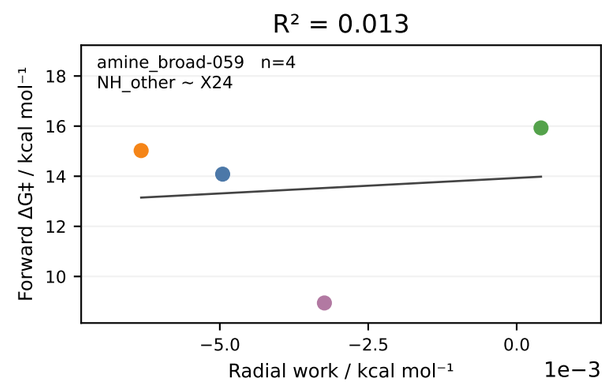
MoriPF:O22···H49; R14:O22···H47
R15:O22···H47; R16:O22···H47



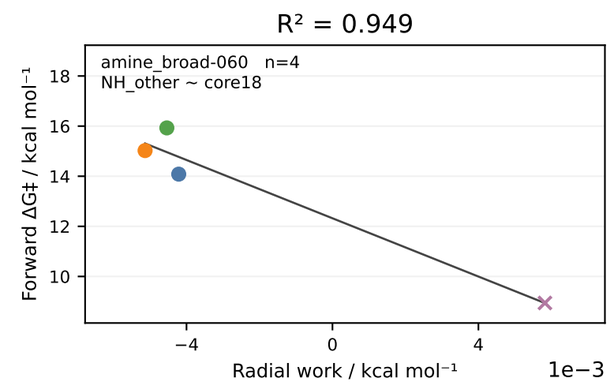
MoriPF:O19···H50; R14:O19···H48
R15:O19···H48; R16:O19···H48



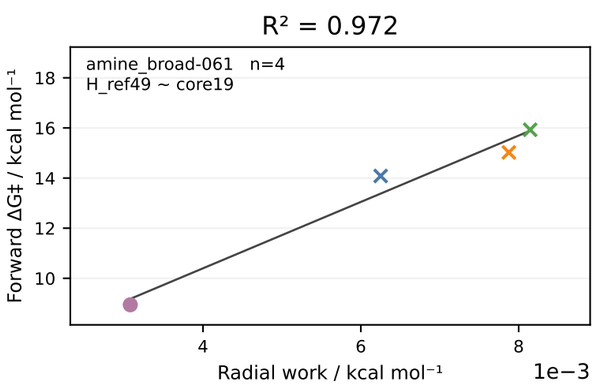
MoriPF:F33···H46; R14:H44···H57
R15:F31···H44; R16:F31···H44



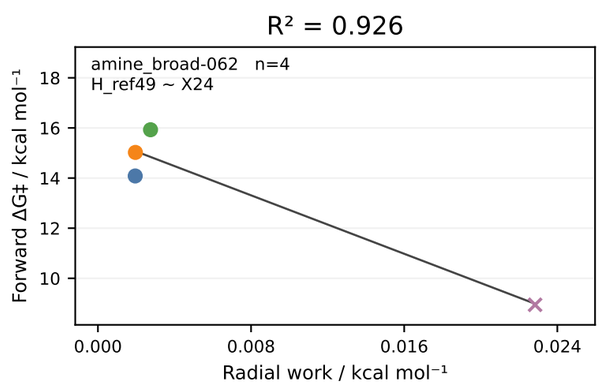
MoriPF:F33···H47; R14:H45···H57
R15:F31···H45; R16:F31···H45



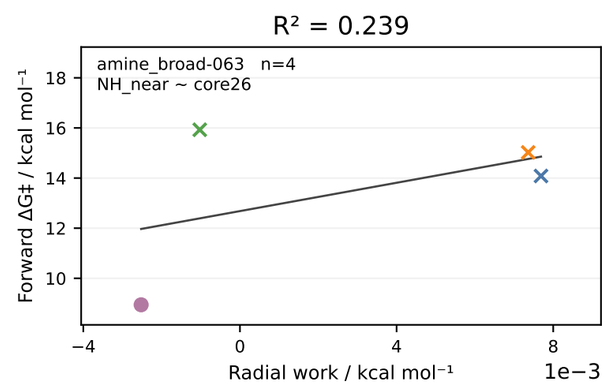
MoriPF:O18···H47; R14:O18···H45
R15:O18···H45; R16:O18···H45



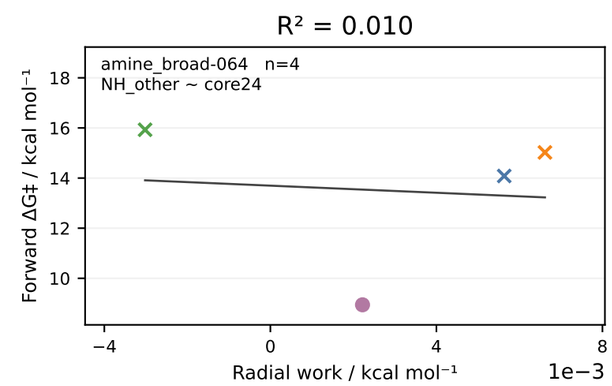
MoriPF:O19···H49; R14:O19···H47
R15:O19···H47; R16:O19···H47



MoriPF:F33···H49; R14:H47···H57
R15:F31···H47; R16:F31···H47



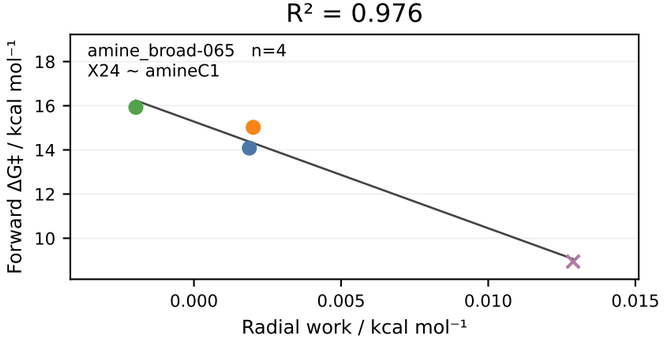
MoriPF:C26···H46; R14:C26···H44
R15:C26···H44; R16:C26···H44



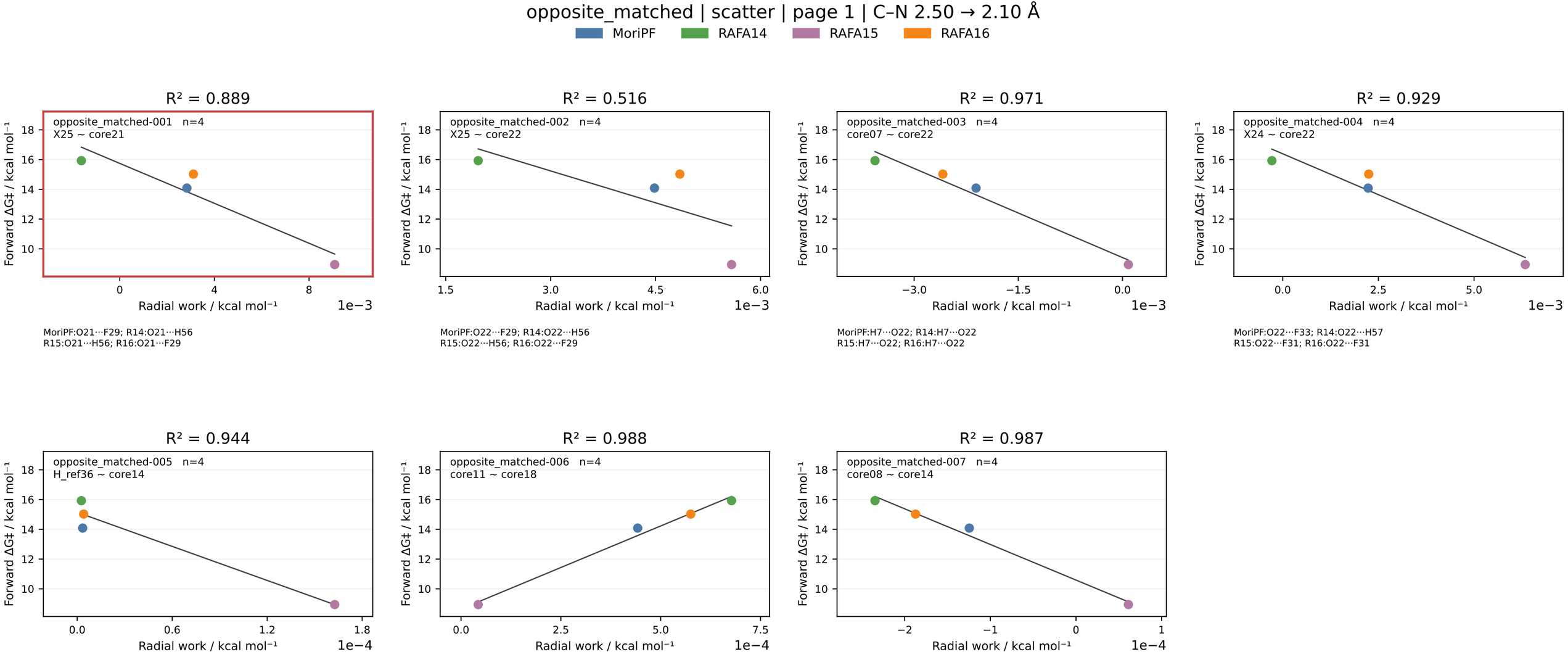
MoriPF:C24···H47; R14:C24···H45
R15:C24···H45; R16:C24···H45

amine_broad | scatter | page 5 | C–N 2.50 → 2.10 Å

MoriPF RAFA14 RAFA15 RAFA16

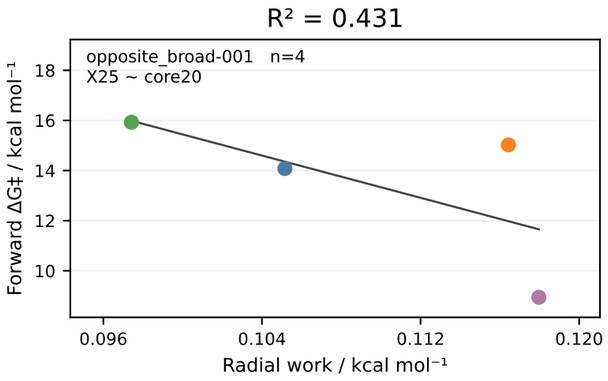


MoriPF:F33···C48; R14:C46···H57
R15:F31···C46; R16:F31···C46

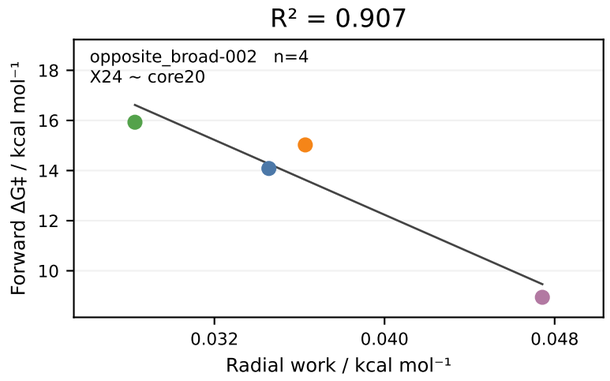

opposite_matched | scatter | page 1 | C–N 2.50 → 2.10 Å
MoriPF
RAFA14
RAFA15
RAFA16
R² = 0.889
opposite_matched-001 n=4
X25 ~ core21
MoriPF:O21···F29; R14:O21···H56
R15:O21···H56; R16:O21···F29
R² = 0.516
opposite_matched-002 n=4
X25 ~ core22
MoriPF:O22···F29; R14:O22···H56
R15:O22···H56; R16:O22···F29
R² = 0.971
opposite_matched-003 n=4
core07 ~ core22
MoriPF:H7···O22; R14:H7···O22
R15:H7···O22; R16:H7···O22
R² = 0.929
opposite_matched-004 n=4
X24 ~ core22
MoriPF:O22···F33; R14:O22···H57
R15:O22···F31; R16:O22···F31
R² = 0.944
opposite_matched-005 n=4
H_ref36 ~ core14
MoriPF:N14···H36; R14:N14···H34
R15:N14···H34; R16:N14···H34
R² = 0.988
opposite_matched-006 n=4
core11 ~ core18
MoriPF:H11···O18; R14:H11···O18
R15:H11···O18; R16:H11···O18
R² = 0.987
opposite_matched-007 n=4
core08 ~ core14
MoriPF:H8···N14; R14:H8···N14
R15:H8···N14; R16:H8···N14
Forward ΔG‡ / kcal mol⁻¹
Radial work / kcal mol⁻¹
1e−3
1e−4

opposite_broad | scatter | page 1 | C–N 2.50 → 2.10 Å

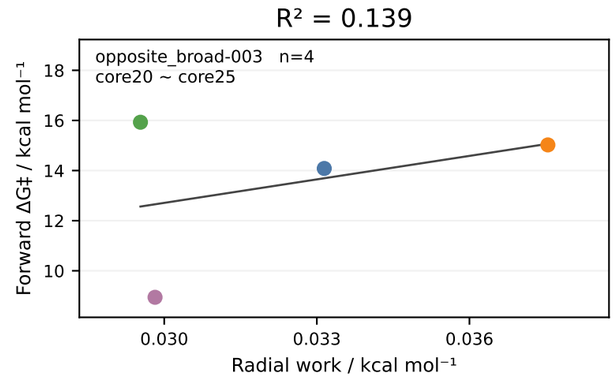
MoriPF RAFA14 RAFA15 RAFA16



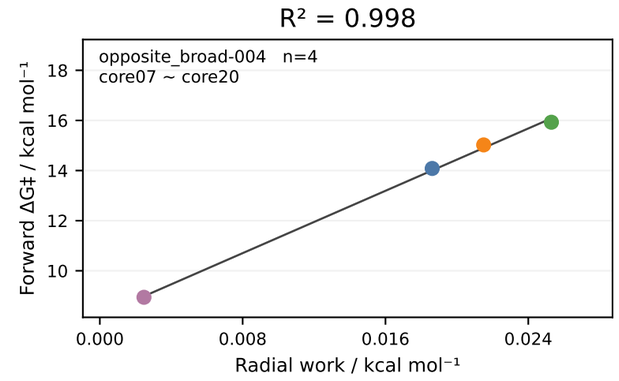
MoriPF:C20···F29; R14:C20···H56
R15:C20···H56; R16:C20···F29



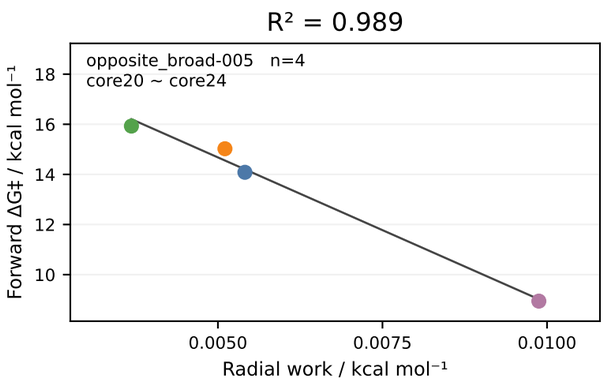
MoriPF:C20···F33; R14:C20···H57
R15:C20···F31; R16:C20···F31



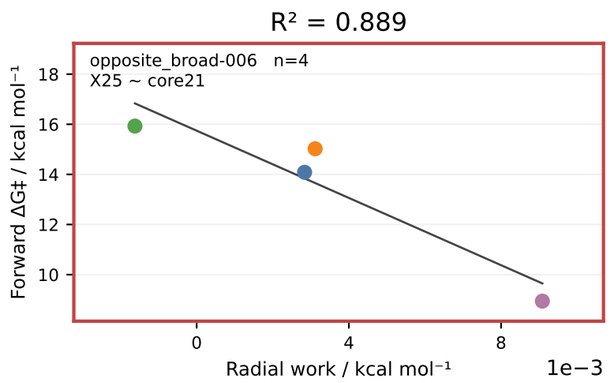
MoriPF:C20···C25; R14:C20···C25
R15:C20···C25; R16:C20···C25



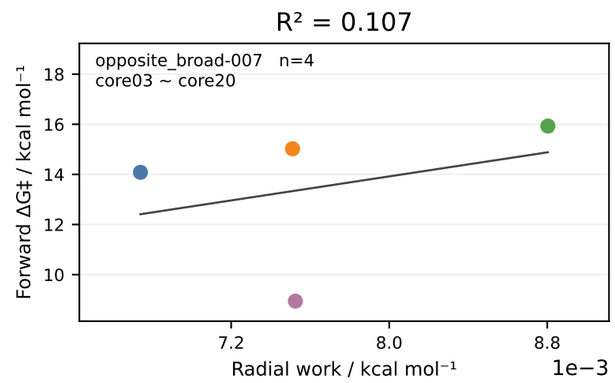
MoriPF:H7···C20; R14:H7···C20
R15:H7···C20; R16:H7···C20



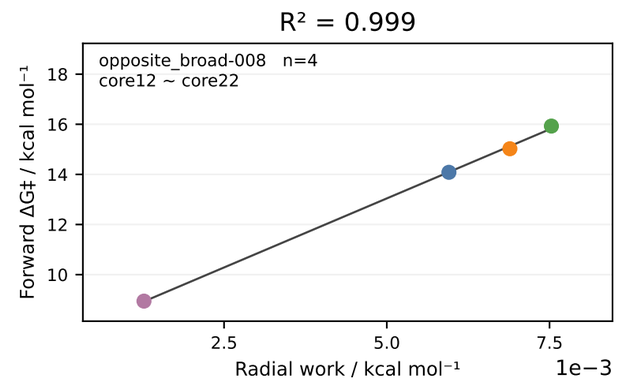
MoriPF:C20···C24; R14:C20···C24
R15:C20···C24; R16:C20···C24



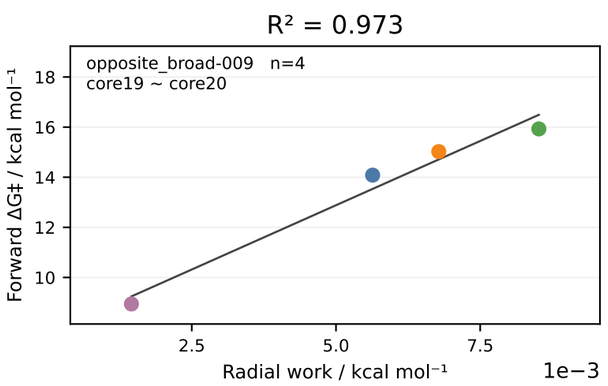
MoriPF:O21···F29; R14:O21···H56
R15:O21···H56; R16:O21···F29



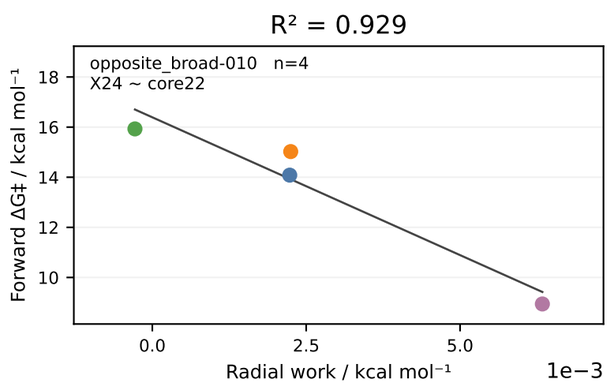
MoriPF:C3···C20; R14:C3···C20
R15:C3···C20; R16:C3···C20



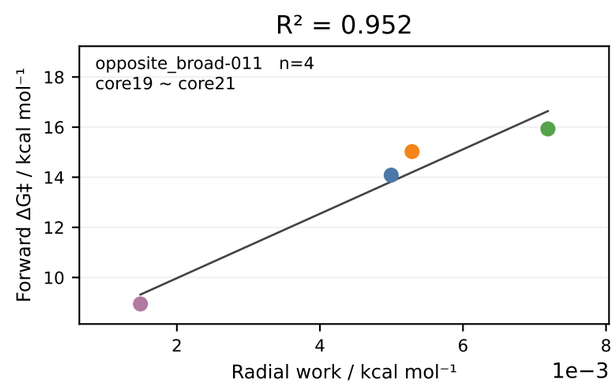
MoriPF:C12···O22; R14:C12···O22
R15:C12···O22; R16:C12···O22



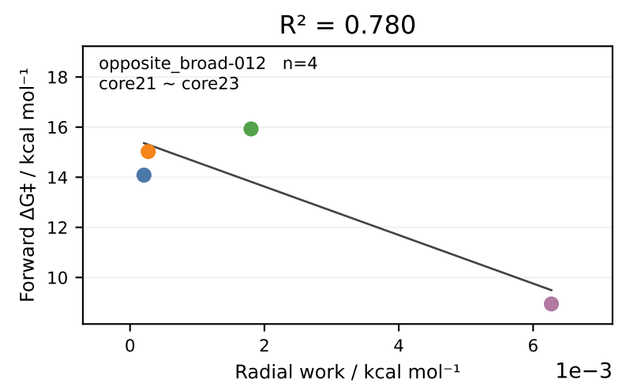
MoriPF:O19···C20; R14:O19···C20
R15:O19···C20; R16:O19···C20



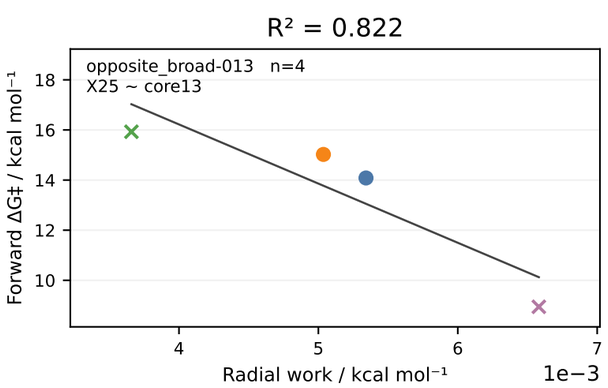
MoriPF:O22···F33; R14:O22···H57
R15:O22···F31; R16:O22···F31



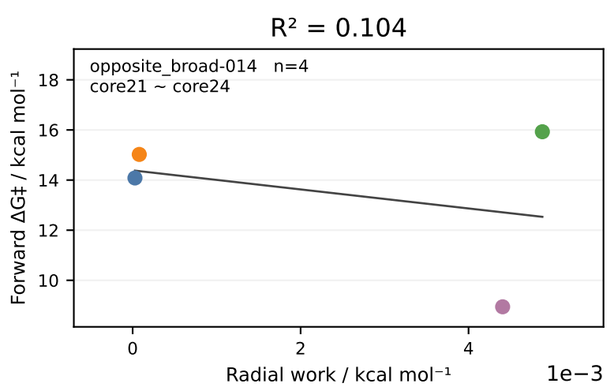
MoriPF:O19···O21; R14:O19···O21
R15:O19···O21; R16:O19···O21



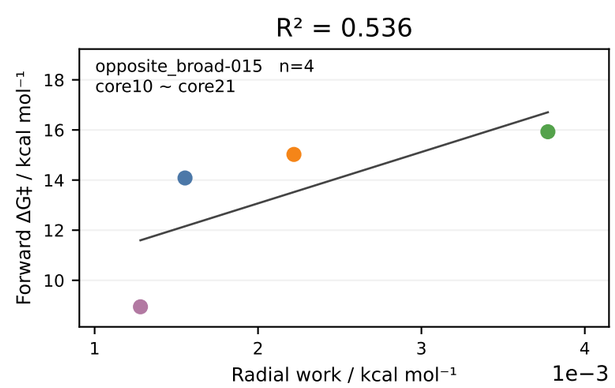
MoriPF:O21···C23; R14:O21···C23
R15:O21···C23; R16:O21···C23



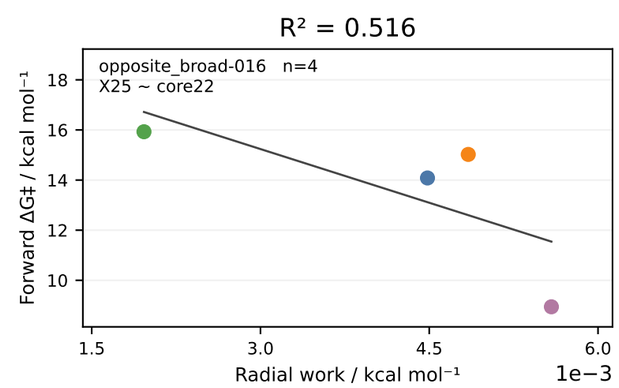
MoriPF:C13···F29; R14:C13···H56
R15:C13···H56; R16:C13···F29



MoriPF:O21···C24; R14:O21···C24
R15:O21···C24; R16:O21···C24

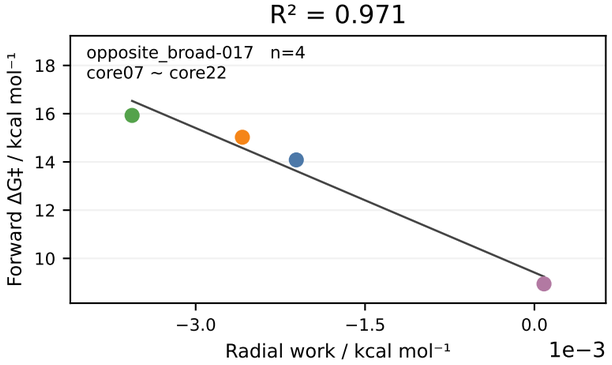


MoriPF:C10···O21; R14:C10···O21
R15:C10···O21; R16:C10···O21

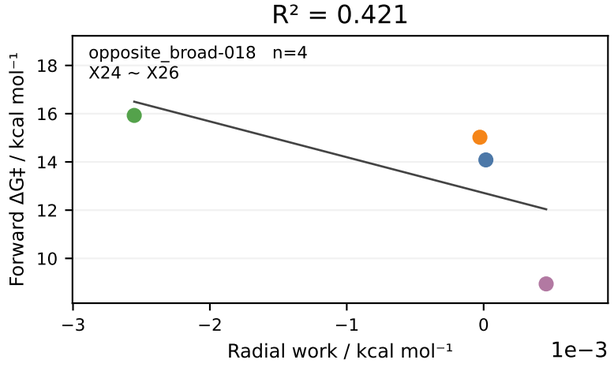


MoriPF:O22···F29; R14:O22···H56
R15:O22···H56; R16:O22···F29

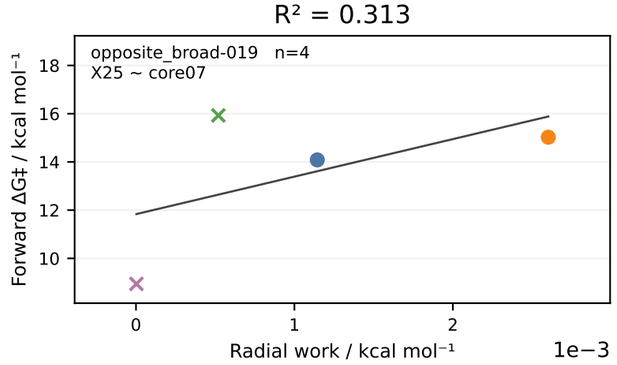
MoriPF RAFA14 RAFA15 RAFA16



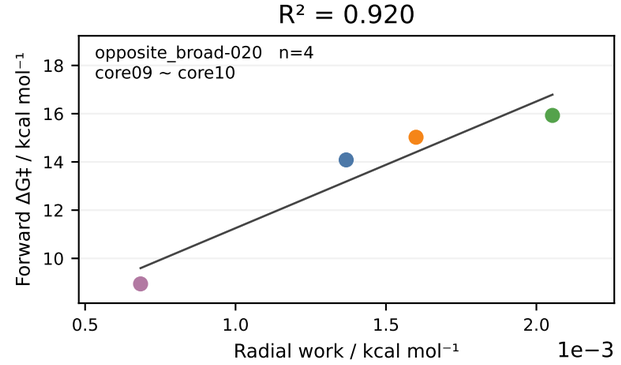
MoriPF:H7···O22; R14:H7···O22
R15:H7···O22; R16:H7···O22



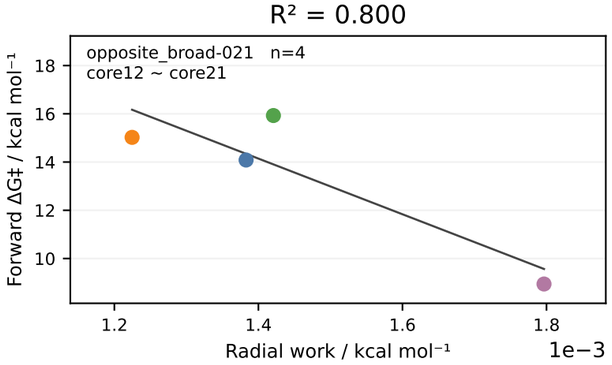
MoriPF:F32···F33; R14:F31···H57
R15:F30···F31; R16:F31···H56



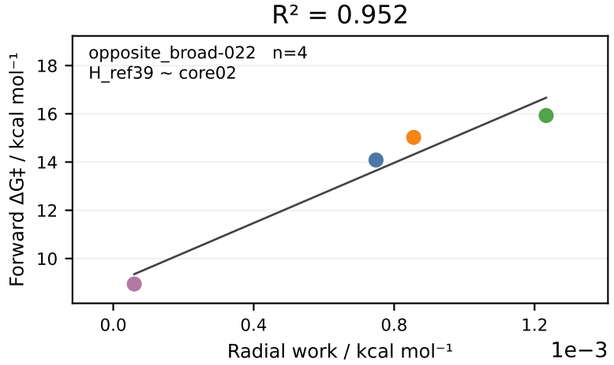
MoriPF:H7···F29; R14:H7···H56
R15:H7···H56; R16:H7···F29



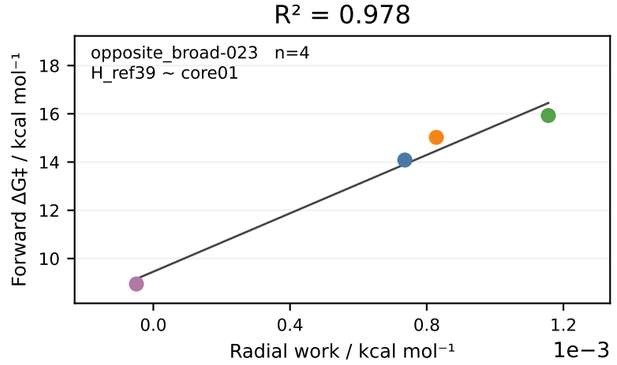
MoriPF:H9···C10; R14:H9···C10
R15:H9···C10; R16:H9···C10



MoriPF:C12···O21; R14:C12···O21
R15:C12···O21; R16:C12···O21



MoriPF:C2···H39; R14:C2···H37
R15:C2···H37; R16:C2···H37



MoriPF:C1···H39; R14:C1···H37
R15:C1···H37; R16:C1···H37